



Federal Office
for Information Security

5G Risk Analysis

Detailed Analysis: **Radio Access Network (RAN)**



Version History

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1 List of Abbreviations

3GPP	3rd Generation Partnership Project
AAU	Active Antenna Unit
AM	Alarm Management
AMF	Access and Mobility Management Function
AS	Access Stratum
BIOS	Basic Input/Output System
CB	Cell Broadcast
CGI	Common Gateway Interface
CIA	Confidentiality, Integrity, Availability
CLI	Command Line Interface
CM	Configuration Management
CORBA	Common Object Request Broker Architecture
CP	Control Plane
CU	Central Unit
DoS	Denial of Service
DU	Distributed Unit
eNB	eNodeB (4G base station)
E-UTRA	Evolved Universal Terrestrial Radio Access
F1-U	F1 User plane interface
F1-C	F1 Control plane interface
ng-eNB	ng-eNodeB (dual connectivity base station: 4G protocol towards UE, 5G interfaces towards Core)
gNB	gNodeB (5G base station)
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GTP	GPRS Tunnelling Protocol
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ICMP	Internet Control Message Protocol
IIS	Internet Information Services
IMS	IP Multimedia Subsystem
IoT	Internet of Things
IP	Internet Protocol
IT	Information Technology
LMP	Local Maintenance Port
MAC	Media Access Control

MEC	Multi EDGE Cloud
remote MGMT	remote Management Interface
MHA	Mast Head Amplifier
MIB	Master Information Block
MNO	Mobile Network Operator
MTBF	Mean Time Between Failures
NAS	Non-Access Stratum
NG RAN	Next Generation Radio Access Network
NMS	Network Management System
NR	5G New Radio
O&M / OAM	Operations and Maintenance
PASTA	Process for Attack Simulation and Threat Analysis
PDCCP	Packet Data Convergence Protocol
PDU	Protocol Data Unit
PM	Performance Management
PWS	Public Warning System
RA	Risk Analysis
RAN	Radio Access Network
RLC	Radio Link Control
RRC	Radio Resource Control
RU	Radio Unit
SDAP	Service Data Adaptation Protocol
SFTP	Secure File Transfer Protocol
SIB	System Information Block
SMF	Session Management Function
SNMP	Simple Network Management Protocol
SRB	Signal Radio Bearer
SSH	Secure Shell Protocol
SSI	Server Side Includes
SUPI	Subscription Permanent Identifier
SYNC	Synchronization Interface
TARA	Threat and Risk Analysis
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UE	User Equipment
UID	User Identifier (UNIX)
UP	User Plane
UPF	User Plane Function
USB	Universal Serial Bus

Uu	5G New Radio Air Interface
VNF	Virtual Network Functions
Xn-C	Xn-Control plane interface
Xn-U	Xn-User plane interface

2 Introduction and Purpose

2.1 Intention

The intention of the document is to dive deep into the analysis of risk scenarios where the **Radio Access Network (RAN)** is the attacked or failing subsystem.

This document shows the current view on the probability that any risk scenarios could become reality.

The study is only available in English language, as the target group is primarily the international IT security community. The study is also in the context of English publications on this topic.

2.2 Link to Framework Document

This document is linked to the framework document and provides relevant information about the **5G Radio Access Network (RAN)** and a probability assessment of relevant risk scenarios described in the framework document. The overall risk assessment for each risk scenario consists of the results of this analysis and the impact described in the framework document. The overall risk assessment for the Radio Access Network (RAN) is summarized in Chapter 6 of this document and in the framework document.

3 Methodology

The methodology used for the 5G Risk Analysis (5G-RA) is based on PASTA. In this document and the related documents, we will refer to the methodology as 5G-RA.

This chapter explains the seven steps of the 5G-RA methodology and assigns them to the corresponding PASTA steps.

Step 1: Definition of Risk Scenarios

In the first step of the 5G-TARA, the 5G network needs to be analysed from the following three perspectives:

1. Public Mobile Network Operator (MNO) Perspective,
2. Private 5G Network Operator Perspective,
3. User Perspective.

Several risks with a significant impact on the society or public safety have been identified by the legislative in the past. To mitigate these risks, laws were set into force, e.g. the legislative to ensure emergency calls are available (TKG §165). Most of these laws define fines for mobile network operators if the regulative obligation is not followed.

These perspectives are explained in detail in Chapter 6 of the framework document. The goal is to identify the possible risks that the 5G network might be exposed to from each perspective.

Step 2: 5G System Diagram

In step 2, a detailed system diagram of the parts of a 5G network needs to be created. This diagram will serve as a basis for the following steps.

Step 3: Refinement of Detailed 5G Subsystem Diagrams

In this step, more technical definitions and explanations of the relevant 5G subsystems are introduced. For example, various network functions are part of the 5G core network. This level of detail serves as a basis for the following steps.

Step 4: Collection of Known 5G Threats

In this step, publicly known 5G specific threats are collected. For this purpose, relevant studies and input documents are collected. This is the basis for the identification of threats in the next step.

Step 5: Search of 5G Vulnerabilities

Step 5 includes the analysis and identification of relevant threats for each risk scenario. The findings of this step are part of the corresponding Detailed Risk Analyses.

Step 6: Examination of Identified Threats

In this step, all threats that were found in step 5 are examined on whether there is a possibility for this threat to be realized. Hypothetical scenarios are used as well as references to successful attacks, which were performed in the past.

Step 7: Risk Calculation

The last step of the methodology is a calculation of the risk for all threats that were identified in step 5 and are still considered relevant after step 6 was carried out. The overall results are presented in Chapter 6 of this document.

The following table assigns the steps of the 5G-RA to the steps of PASTA and gives an overview of the document containing the results of each step.

Table 1: Assignment of 5G-RA steps to PASTA-steps

#	5G-RA methodology	Document Layer	PASTA
1	Definition of Risk Scenarios	Framework	Define Objectives
2	5G System Diagram	Framework	Define Technical Scope
3	Refinement of Detailed 5G Subsystem Diagrams	Framework & Detailed Risk Analysis	Application Decomposition
4	Collection of Known 5G Threats	Detailed Risk Analysis	Threat Analysis
5	Search of 5G Vulnerabilities	Detailed Risk Analysis	Vulnerability and Weakness Analysis
6	Examination of the Identified Threats	Detailed Risk Analysis	Attack Modelling
7	Risk Calculation	Framework & Detailed Risk Analysis	Risk and Impact Analysis

4 5G Network Model: Radio Access Network

The 5G network model introduced in the following sections is used to split the 5G-RA into multiple Detailed Risk Analyses. The following model follows standards (like 3GPP standards) as much as possible. The 5G network model comprises a variety of interconnected subsystems that are categorized on a high-level view as shown in Figure 1. In the context of this risk analysis, a 5G Standalone system is assumed. Therefore, the Radio Access Network (RAN) as well as the Core are 5G-specific subsystems.

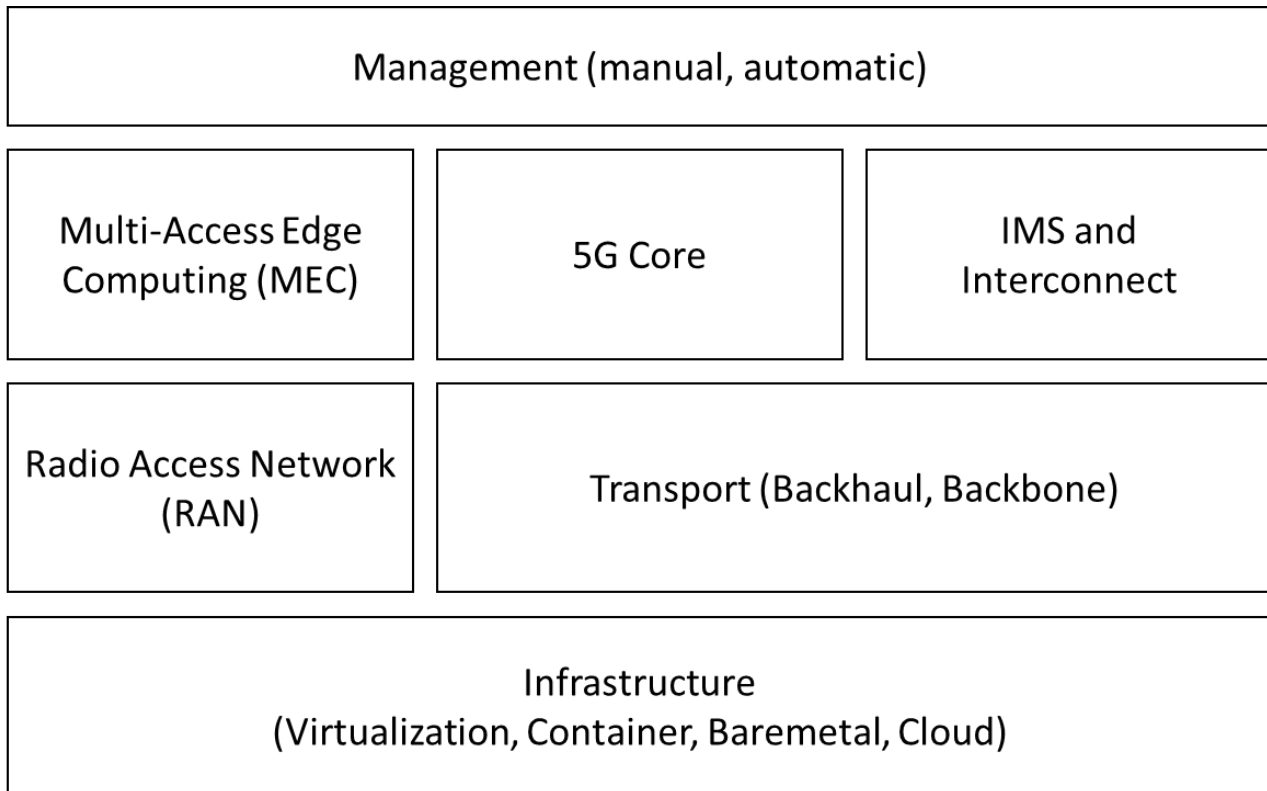


Figure 1: 5G Network Model: High level view

Other subsystems are not specifically designed for the 5G architecture and used either to operate the 5G system or to connect it to other internal and external networks.

The architecture of the 5G Radio Access Network (RAN) is defined by 3GPP in TS 38.401[78], while the 5G RAN in general is specified in 3GPP TS 38.300 [2]. Further descriptions of specific interfaces of the 5G RAN can be found in detailed specifications. This section gives a short overview on the 5G RAN architecture, for more details see [89]. The specific architecture of Open RAN is not in the scope of this document. For a risk analysis on Open RAN as defined by O-RAN Alliance, see [84].

Figure 2 shows a typical deployment of 5G base stations (gNB) with a radio interface (Uu) that provides connectivity for mobile devices. The gNB itself consists of a Central Unit (CU) for high layer processing, one or many Distributed Units (DU) for radio data processing and one or more Radio Units (RU). The RU is not explicitly specified by 3GPP. The 5G RAN may contain 4G base stations, ng-eNodeBs (ng-eNB), for dual connectivity via Evolved Universal Terrestrial Radio Access (E-UTRA) [78]. Interfaces to the RAN management functionality as the remote Management interface (MGMT) and the Local Maintenance Port (LMP) are also considered in the RAN subsystem.

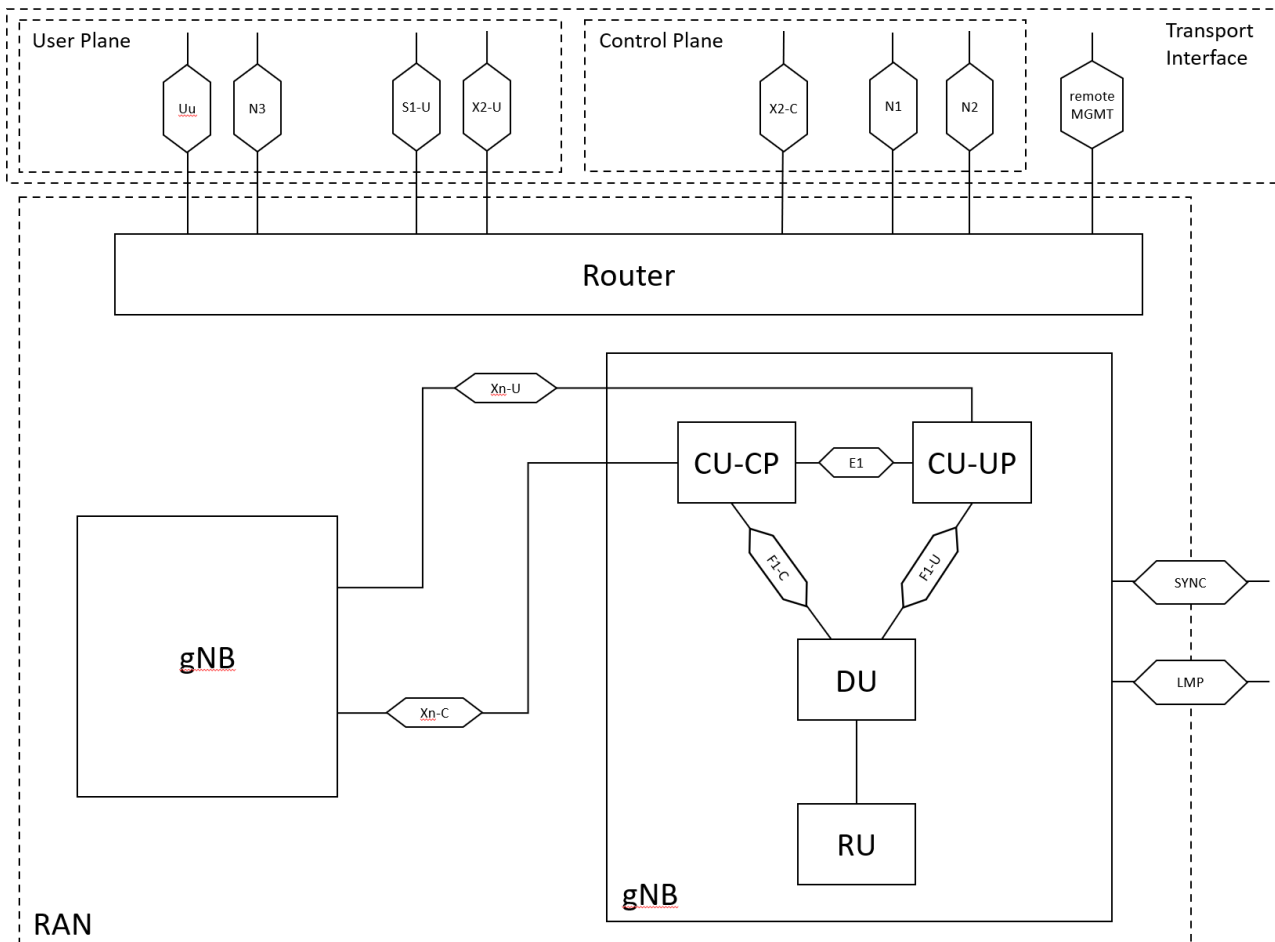


Figure 2: Radio Access Network with gNBs

4.1 Central Unit (Control and User Plane)

The Central Unit (CU) processes the higher layers of the protocol stack. It is specified by 3GPP in TS 38.401 [78]. The CU may be implemented on bare metal or as Virtual Network Functions (VNF) on a server or in a cloud environment.

The CU is separated in Control Plane (CU-CP) and User Plane (CU-UP).

The CU-CP handles the control plane protocols Radio Resource Control (RRC) and Packet Data Convergence Protocol – Control Plane (PDCP-C) of the radio access processing. It can be connected to one or more CU-UPs. The CU-UP handles the user plane protocols Service Data Adaptation Protocol (SDAP) and Packet Data Convergence Protocol – User Plane (PDCP-U) of the radio access processing.

4.2 Distributed Unit

The Distributed Unit (DU) hosts the lower layers of the protocol stack, such as Radio Link Control (RLC), Media Access Control (MAC) and the physical layer. It provides baseband processing and RF functions. It is specified by 3GPP in TS 38.401 [78]. A DU can be connected to several CU-UPs, but only to one CU-CP.

4.3 Radio Unit

The Radio Unit (RU) transmits and receives 5G New Radio (NR) signals. It is not specified by 3GPP. The RU may be implemented as a separated module that is connected via coaxial cables to a passive antenna system. At higher frequencies the RU may be implemented as an Active Antenna Unit (AAU) that consists of a high number of power amplifiers, receivers and dipoles (e.g. 32, 64 or 128) and could actively and adaptively form radio beams.

4.4 Router

Most gNB deployments use an IP router that may provide functionalities like firewall, redundant transport connection and control functions for supplementary hardware as air condition, continuous power supply, etc.

4.5 gNodeB

The gNB consists of a CU-CP and one or multiple CU-UPs, one or multiple DUs and one or multiple RUs.

There are many different implementation scenarios, e.g.:

- CU-CP, CU-UP could be implemented as VNFs e.g. in the cloud or at Multi-Access EDGE Computing locations, whereas DU, RU entities could be spread over different locations.
- CU-CP, CU-UP could be implemented on bare metal at centralized locations, whereas DU, RU entities could be spread over different locations.
- CU-CP, CU-UP, DU, RU entities could be implemented as separate modules at the same location, but at different positions, e.g. RU could be mounted near the antenna and CU-CP, CU-UP, DU could be placed in a rack within a housing.
- CU-CP, CU-UP, DU, RU could be integrated into one device.

4.6 Interfaces

4.6.1 Radio Interface Uu

The User Equipments (UEs) exchange control and user data with the gNBs over the 5G NR radio interface (Uu).

4.6.2 Interfaces E1, F1-C and F1-U

E1 is the interface between CU-CP and CU-UP and is standardized in 3GPP TS 38.460 [80][79][79][79]. This interface is only used for the exchange of data between CU-CP and CU-UP. The E1 Application Protocol Procedures via E1 either are interface management or bearer context management procedures.

The interface F1 specified in 3GPP TS 38.470 [81][80][80] connects CU and DU. As CP and UP traffic is handled separately in CU, F1-C is the connection for the Control Plane traffic, while F1-U is for the User Plane traffic (cf. 3GPP TS 38.472 [82][81][81] and 3GPP TS 38.474 [83]). As the functionality of the F1 interface is clearly defined in 3GPP TS 38.470 and 3GPP TS 38.472, the operator has the possibility to deploy CUs and DUs supplied by different vendors.

4.6.3 Interfaces N1, N2 and N3

The interfaces N1, N2, and N3 all connect to functions of the 5G Core.

N1 is the interface between UE and AMF. The N1 interface connects UE and core network via the Non-Access Stratum (NAS). This interface is transparent to the RAN.

N2 is the interface between RAN and AMF. A base station can be connected to several AMFs. RAN and AMF transmit control messages between each other via the N2 interface. The interfaces N1 and N2 are also named NG-C. The signaling is handled via the NG Application Protocol as specified in 3GPP TS 38.413.

N3 is the interface between RAN and UPF. It is also called NG-U and specified in 3GPP TS 38.415. It connects RAN and UPF for the transmission of user information. A gNB can be connected to several UPFs.

4.6.4 Interfaces Xn-C and Xn-U

The Xn interface is specified by 3GPP in TS 38.420 [79]. It connects two gNodeBs with each other. The control plane interfaces Xn-C and the user plane interface Xn-U provide connection between adjacent gNBs.

3GPP TS 38.423 specifies the Xn Application Protocol for the control plane signaling on interface Xn-C:

- Global procedures of Interface management and error handling as setup, configuration update, cell activation, reset, error indication and removal,
- Mobility management as handover procedures, sequence number status transfer, UE context retrieval and release and RAN paging and
- Dual connectivity functions as secondary node addition, reconfiguration, modification, release.

The user plane interface Xn-U transmits application data for e.g. data forwarding during handover or while using dual connectivity between gNBs as defined in 3GPP TS 38.425. The data is transferred via GTP-U tunnels.

4.6.5 Interfaces S1-U, X2-C and X2-U

The interface X2 is specified by 3GPP in TS 36.420 and consists of X2-C for the Control plane traffic and X2-U for the User plane traffic. It connects a gNB with an eNB as well as two eNBs. It is therefore not relevant for standalone mode. Nevertheless X2 needs to be considered as an entry point of attack even in the case if it is not in use. It is separated in control plane signaling via X2-C and User Plane signaling via X2-U.

The **S1-U** interface connects RAN to the Evolved Packet Core (EPC) in non-standalone mode. In standalone mode, the NG interfaces N2 and N3 are used instead. Nevertheless S1-U needs to be considered as an entry point of attack even in the case if it is not in use.

4.6.6 Management Interfaces

There are different implementation scenarios for the interfaces for orchestration and management of the RAN subsystem. For simplification purposes that kind of interfaces are named remote management interfaces (remote MGMT) and Local Maintenance Port (LMP) within this document.

The management functionalities include:

- Configuration Management (CM) of RAN parameters covers the initial parameter settings of new gNBs, activation or adoption of RAN functionalities and optimization of operations. Typically, CM changes are done automatically or semi-automatically via the Network Management System (NMS)

by third-party parameter planning or optimization tools. CM may also be done manually via remote MGMT or LMP.

- **Alarm management (AM)**
Most RAN nodes provide interfaces to external alarming sources. Alarms are notified at the network control center of the MNO.
Typical external alarms are:
 - Temperature alarm,
 - Main power supply alarm,
 - Continuous power supply active,
 - Door open alarm and
 - Motion alarm.
- **Performance management (PM)** functions deliver information about network performance as call quality, call drop rate, data throughput. PM raw data is continually collected and aggregated within the gNB and transferred in pre-defined intervals (e.g. every 15 or 60 minutes) to the Network Management System (NMS). PM data may be used together with the CM functionality for automatic parameter optimization.
- New network element software releases can be transferred to the gNB remotely or locally via LMP. Software management often provides functionalities to fall back to previous software loads.
- **Management of third-party equipment:**
gNBs often deliver functionalities to manage third-party equipment locally e.g. via a network element manager or remotely. Typical examples are the management of remote electronic tilts as initializing tilt device, setting desired tilt value or of Mast Head Amplifiers (MHA). MHAs are used to amplify the receiving signals. Third-party equipment management functions may control the amplification value or report the power consumption of the MHA device.

Services include:

- A **webserver** for gNB's user interface (CM, AM, PM, software management, third party equipment management) provides the user interface to remote or local users to manage various functions that are relevant to integration, maintenance and troubleshooting of gNBs.
- **Exchange of CM, AM and PM data** with network management system can be done via e.g. CORBA or XML file transfer. Efficient management of big networks can only be done via highly automated tools as CM, AM and PM.
- Management services, that are controlled by network management system e.g. for downloading and activating gNB software loads or for controlling the states gNB (locking / unlocking of gNB/cells)
A practical way to upgrade gNB software loads is to download the software remotely and activate it.
- A **Command line interface (CLI)** can be used for the management of gNB on an operating system level.
- **Secure File Transfer Protocol (SFTP)** may be used for the exchange of cryptographic.
- **Simple Network Management Protocol (SNMP)** is usually used for management of IP routers which provide connectivity at gNB sites.
- **Secure Shell Protocol (SSH)** provides secure communications that are used e.g. for remote login and CLI.
- **Hypertext Transfer Protocol Secure (HTTPS)** is used for secure communication between gNB's webserver and user's web browser.

Interface technology:

- Transport Layer Security (TLS)
- LMP: RJ45 (Ethernet/IP), USB mass storage, USB human interface devices as keyboard and mouse
- Remote MGMT: IP transport interface of gNB via fiber or microwave

4.6.7 Interface for Synchronization with Global Navigation Satellite System Receiver

In most networks the majority of gNBs get the required network timing information through the transport interface via timing over packet service or other solutions. But in some deployment scenarios a Global Navigation Satellite System (GNSS) receiver (e.g. GPS) is used instead, which is more expensive per gNB but delivers precise synchronization data. GNSS receivers may be connected to a synchronization interface (SYNC) of the gNB.

4.6.8 Transport Interface

The physical connection between gNB and the core or other gNBs is often called Transport Interface. It carries the control plane and the user plane as well as other services (e.g. remote MGMT).

The transport interface is normally implemented as fiber or copper ethernet. In special cases the transport interface may also be implemented as directional radio/ microwave transmission.

5 Risk Scenarios

This chapter provides the current view on a selected set of risk scenarios. It deals with an estimation of probabilities that either of these risk scenarios could become reality. These estimations are based on detailed discussions of potential scenarios of weaknesses or attacks. These discussions conclude with an assessment of the probability of occurrence.

The following table shows whether the RAN subsystem is a relevant attack domain for either of the risk scenarios. The risk scenarios themselves are further described in the framework document.

Perspective	Section	Risk Scenario	RAN	CIA
Public MNO Perspective	5.2.1	Emergency Call (TKG § 164)		A
	5.2.2	Cell Broadcast (TKG § 164a)	x	A
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	x	C
	5.2.4	Emergency Preparedness (TKG § 184)	x	A
	5.2.5	Regional Network Failure	x	A
	5.2.6	Multi-Regional Network Failure		A
	5.2.7	Failure of Mobile Service		A
	5.2.8	Fraud		I
	5.2.9	Subscriber-Data Leak	x	C
User Perspective	5.3.1	Urgent Telephone Calls Not Possible	x	A
	5.3.2	Authenticity of the Receiver of a Call		I
	5.3.3	Integrity of Transmitted Data	x	I
	5.3.4	User's Identity	x	C
	5.3.5	Eavesdropping of a Communication	x	C
	5.3.6	Slowing Down/Restricting the Connection	x	A
	5.3.7	Connection Failure	x	A
	5.3.8	Tapping or Altering of the Geographical Position	x	C
Private 5G Network Operator Perspective	5.4.1	Network Failure	x	A
	5.4.2	Degraded Services/Connections	x	A
	5.4.3	Information Disclosure	x	C
	5.4.4	User Identities (Authenticity/ Integrity)	x	I

5.1 Attack Paths and Risk Sources

For either of the risk scenarios a couple of different attack paths and risk sources will be discussed within this document. The result of the discussion will be a probability level for the impact defined in the respective risk scenario.

The term "attack path" refers to a sequence of steps or actions that an attacker can take to exploit vulnerabilities in a system or network, leading to security issues. It outlines the route or method an attacker may follow, often involving a series of interconnected actions that exploit weaknesses inherent in the system.

A risk source is defined as the origin or cause of a potential risk or threat. A risk source is identified as the factor or element that can lead to vulnerabilities or hazards.

As the attack paths and risk sources often exhibit similarities for several of the risk scenarios, each attack path and risk source will be introduced from a RAN perspective in this chapter.

In general, an attacker with legitimate access to MNO's assets that shall be protected is called "internal attacker". An internal attacker could have access to physical or cryptographic keys, to buildings or rooms or to networks or systems.

5.1.1 Misconfiguration (AP1)

Misconfiguration as a possible attack path [21] concerns all configurable parts of a system. A legitimate configuration can either be deployed by employees of the MNO or the manufacturer or by configuration tools. The primary cause of misconfiguration is considered here as by human error thus happening unintentionally, but intentional misconfiguration, e.g. by an internal attacker or by a former employee still having access [53], is not precluded.

Misconfiguration may also be the result of other attack paths, e.g. implementation flaws may result in the use of outdated configuration files, or insufficient access control may enable an attacker to gain access to a local configuration interface or to the central network management system. Such attack paths are considered as implementation flaws and insufficient access control, respectively, and not considered as misconfigurations.

Misconfigurations may persist undetected if the impacted functionality is used only sporadically (e.g. public warning messages) or if configuration or security impacts of that functionality are not frequently monitored.

The 5G air and network interfaces provide a large degree of configurability such as waveform numerology, signal parameters in time and frequency domain, e.g., the presence and density of reference signals, radio resources and transmission properties configured for control signaling, parameters for unified access control or handover configuration, use of encryption and integrity protection, use of carrier aggregation or dual connectivity, use of filtering and replay protection, etc. [2]. Further, the 5G RAN includes a variety of proprietary algorithms that may be configured to some extent by the MNO such as radio admission control handling the access of users to the RAN, or radio resource management including packet scheduling, link adaptation and slicing, the selection and use of multiple antenna transmission/reception schemes etc. The network management system includes domains for the different RAN network elements, Remote Unit (RU), Distributed Unit (DU), and Centralized Unit (CU), that may be co-sited or deployed at separate sites, as well as the different network interfaces and respective routing or switching equipment. This implies that the RAN configuration may be under the control of multiple teams or individuals, adding further complexity to the RAN management.

5.1.1.1 Confidentiality (AP1-C)

User plane ciphering may be disabled by misconfiguration on the air interface and/or network interfaces, thus enabling an attacker to eavesdrop on unencrypted traffic on the air interface and/or the network interfaces, e.g., midhaul or backhaul interfaces [3][28].

5.1.1.2 Integrity (AP1-I)

User plane integrity protection may be disabled by misconfiguration on the air interface and/or network interfaces, thus enabling an attacker to tamper traffic on the air interface and/or the network interfaces, e.g., midhaul or backhaul interfaces [3][28].

5.1.1.3 Availability (AP1-A)

Misconfiguration of the 5G RAN may disable one or multiple radio cells or sites, but due to the complexity of the 5G RAN, misconfiguration will more likely lead to degraded network performance. Examples for RAN misconfigurations include:

- Misconfiguration of air interface parameters including those transmitted within the Master Information Block (MIB) or within one of the System Information Blocks (SIB) can lead to disabled cells, e.g., by indicating cell barring on the MIB [2][39], or lead to degraded cells, e.g., by flawed configurations of radio resource management algorithms, handover or cell selection thresholds, or Radio Resource Control (RRC) state transition timing.
- Misconfiguration of the radio admission control can cause an overload of the cell or of one or multiple RAN slices. The use of slicing in the RAN adds further complexity to properly configure the radio admission control as multiple slices share the same set of radio resources [12][73].
- An improper dependent RRC configuration in a scenario using multiple serving cells can lead to disruption of connectivity and throughput drops [4].
- Misconfiguration of filtering or replay protection may cause congestion on network interfaces, e.g. improper or absent GTP (GPRS Tunnel Protocol) filtering may cause congestion of the NG interface between the RAN and the core network, or improper or missing replay protection may cause congestion of RRC signaling or the NG interface [13].

5.1.2 Insufficient Access Controls (AP2)

Generally, there are two types of access methods: physical access and software-based access. A main difference between physical and software-based access is, that after accessing the RAN components physically, further action and know-how is needed to access and manipulate the software running on the hardware.

Physical access: A RAN system is composed of different hardware elements that are hosted at the radio sites or in data centers (e.g. a centralized unit). The respective location should have access controls so that it can generally only be accessed by authorized personnel.

Insufficient physical access protection at the respective sites may enable an attacker to gain local access to respective hardware elements that comprise different types of interfaces for traffic and administration [22]. The attacker has then multiple options to utilize the access to the system. User and management traffic can be accessed, the RAN or its components can be reconfigured to achieve an insecure configuration, service can be disrupted, or hardware can be destructed [72][73]. It should also be borne in mind that additional access controls may be set up for actual access to the configuration interfaces. In this case, weak or default credentials would weaken security.

The 5G trust model assumes that trust within the network is considered as decreasing the further one moves from the core [58]. This implies that the radio sites at the edge of a network can generally be considered as less well protected than the more central sites or data centers [34].

Software-based access: A RAN system comprises multiple software elements that play together to provide users access to the mobile network. Some software elements are hosted on cell sites, while elements such as Centralized Units (CU) (and also Distributed Units, DU) can be hosted in the cloud. The respective cloud infrastructures can be hosted in a data center in the adjacent area to meet e.g. latency requirements.

As the cloudification of RAN components is becoming more important, cloud providers are gaining more and more importance in the infrastructure ecosystem [1].

To manage the RAN systems, authorized employees have virtual access to the management network and management interfaces, as well as to the data in the respective systems. Secure design principles are very important in this case, as insufficient access control to these interfaces and networks can affect the RAN over a larger area when compromised by an attacker.

Another important attack strategy to be considered is to find unprotected shared resources. Shared resources that do not have appropriate access control mechanisms in place could also be used to gain access to sensitive user or operational information [54]. Examples for shared resources are storage area networks (SANs), AWS S3 instances, shared data between instances on the same hypervisor or shared memory or CPUs.

5.1.2.1 Confidentiality (AP2-C)

Insufficient access controls can enable an attacker to access confidential material like business secrets or other sensitive information like private user information or communication data [23]. This includes also insufficiently protected shared resources. Based on the extracted data, an attacker can then utilize the information. One example is the compromise of credentials that can allow vertical privilege escalation and the access to more critical components like management networks.

5.1.2.2 Integrity (AP2-I)

If an attacker can gain access to insufficiently protected resources, the respective data can be destroyed or manipulated even prior to transmission and affect business processes or user communication allowing to inject malicious payloads or sensitive information [55].

When targeting management interfaces or systems, an attacker can reconfigure the system to disable integrity protection enabling to utilize further attack paths.

5.1.2.3 Availability (AP2-A)

With access to the RAN hardware service disruptions can be achieved through cable cuts, theft of cables or other components [34], arson, or other types of vandalism [56]. Another way to disrupt services can be achieved through the access to hardware ports and interfaces that are typically used for management and reconfiguration. If not sufficiently protected, an attacker can access the system's software and put it into an unstable operating state.

An attacker with access to management interfaces or systems can reconfigure the system [57]. One attack could disable one or several radio cells for normal users by misconfiguring the cell(s) into barred state.

5.1.3 Implementation Flaw (AP3)

Implementation flaws of hardware, software and interfaces can lead to severe security issues.

Software and hardware testing is essential to reach the desired functionality and quality of a product. The increase of the complexity of products can be challenging when it comes to achieve a sufficient test coverage. Therefore, some important test cases might not be considered, or some effects related to rare events might remain undetected due to missing tests. This is also important when it comes to software updates. If updates are not backwards compatible or the effects of the update are not well tested, this can lead to problems with availability in particular [10].

As the RAN software is delivered by a software vendor, the vendor is also responsible for providing updates that also includes improvements of software security, stability and performance. The security of software depends on multiple factors such as secure development, security monitoring and deployment lifecycles. Even though the manipulation of development pipelines or external code might be a result of insufficient access control, a supply chain attack is considered as an implementation flaw, as it is caused by insufficient care in the security implementation and deployment processes. The increasing complexity of software stacks and the use of open-source software can therefore pose a challenge for pipeline security, development, deployment and maintenance.

Open-source software projects such as the Linux operation system, cloud platforms like Open Stack or Kubernetes are widely used in data centers and often rely on other open-source software as well. They are typically maintained by specialized experts that built a high level of trust and reputation. Smaller projects often face bigger challenges when it comes to development, maintenance and security as they are managed by a smaller number of people.

Although backdoors could have been identified, as in the recent case of the xz project [14][15], they have often already partially propagated into some systems. Other types of backdoors to be considered are cryptographic backdoors, where cryptographic parameters, routines or key derivation processes are manipulated or weakened.

In recent years, attacks on open-source software and cloud infrastructures have increased [74][75], making it necessary to utilize advanced software testing, security testing and security monitoring to prevent supply chain attacks. It can be expected that such kinds of attacks will increase further.

But not only software-based implementation flaws have to be considered. Also faulty hardware implementations [76] or even manipulated hardware [77][77] has to be considered. Specialized hardware is typically produced by specific vendors. Hardware vendors optimally know each component that is used in their hardware and where those components originated from, but this is not always the case. Further, additional security checks of hardware components are challenging. A malicious supplier of a component could produce and deliver manipulated chips, or alternatively, the vendor could add extra components with malicious intent. Both methods could be used to weaken cryptography, integrate backdoors or to gather and extract information that is flowing inside of the hardware.

5.1.3.1 Confidentiality (AP3-C)

If not properly implemented, products can contain flaws that can lead to exposure of confidential information [13][27][57]. Any actors can then be put in a position to read information that was actually intended for other recipients. This includes faulty allocation of memory, storage or shared resources.

Cryptographic methods and algorithms have to be carefully implemented. Using default or insecure keys can introduce severe security issues. A flaw in the implementation or configuration of cryptographic functions or keys can allow an attacker to gain access to confidential data. In this regard, the internal cryptographic management (e.g. updates to key material and algorithms) has also to be considered.

Flaws in the identity management, such as infrequent updates of temporary user identifiers may enable the tracking of the position of a user.

Furthermore, as supply chain attacks have become more popular over the recent years, in general both compromised software and hardware can allow an attacker to place backdoors and use them to extract confidential information.

5.1.3.2 Integrity (AP3-I)

The integrity of a RAN can be influenced by compromised software and hardware if it is not checked by extensive security checks before delivery. An attacker could use the delivered software or hardware to manipulate data that is used to control the RAN components, e.g. a configuration affecting the integrity protection.

Further, faulty software updates can affect the integrity of data if the updated software is accessing and altering existing data due to (a part of) an update routine.

5.1.3.3 Availability (AP3-A)

The availability of RAN services can be affected by faulty software updates if they have not been sufficiently tested before delivery. If the software of a large number of base stations is updated at once, this can lead to a disruption of services on a larger area and scale [24][25].

Implementation flaws in the management of cryptographic components (e.g. PKI) can also lead to a disruption of services. One example is the expiry of a certificate, which means that components can no longer communicate with each other.

Further, some configuration constellations might not be considered in software and hardware tests, which can lead to service disruptions if the implementations do not contain a corresponding logic [4].

Services based on virtualized network function implementations such as Centralized Unit (CU) or RAN network slices can be degraded or disrupted if they share computing resources claimed by fake computation-intensive tasks through benign or malicious applications (malware) [8].

5.1.4 Specification Weakness (AP4)

The 5G System introduced a new security architecture including elements of a zero-trust architecture where the user equipment does authenticate the network and the network does authenticate the user equipment [68][67]. Some of the known shortcomings of the LTE security architecture, e.g. [28], got corrected for the 5G System. In particular, the support of concealment of the Subscription Permanent Identifier (SUPI) prior to transmission over the air interface got introduced.

While the security architecture of the 5G System is still evolving, some shortcomings related with downgrade attack options or unencrypted pre-authentication messages were reported in the literature, e.g. [3].

Some of the security features of the 5G System can be disabled via configuration, ciphering and integrity protection on the user plane, in particular, as they are only optional [68]. Misconfigured security features are considered as attack paths caused by misconfiguration rather than being caused by specification weaknesses.

The SS7 vulnerability may be cited as a prominent example of a specification weakness that got exploited by attackers to cause damage to mobile communication users [29].

5.1.4.1 Confidentiality (AP4-C)

Confidentiality of the user identity can be compromised by a downgrade attack, where an attacker forces the 5G-capable user equipment to switch to using the LTE air interface (or another legacy technology), e.g., by deploying a fake LTE base station in proximity of the user. Unlike 5G, the LTE air interface does not support concealment of the Subscription Permanent Identifier (SUPI) of the user, which enables an attacker to obtain knowledge of the identifier [3][5]. This attack is also known as IMSI (International Mobile Subscriber

Identity) catching. The fact that downlink information on MIB/SIB is transmitted in unencrypted form simplifies the implementation of the LTE fake base station. Access restrictions to LTE networks can in principle be configured for a user, but this may often lead to coverage issues and is therefore undesirable.

Furthermore, the characteristics of unprotected data sent by the UE could be used to fingerprint user's devices. In particular, information related with the access identity / priority of the user is transmitted unencrypted by the user equipment during the RRC procedures for link establishment [9]. These methods enable an attacker to identify and track the presence of users with priority access such as police forces or rescue teams in a radio cell, as knowledge of the access priority can help to distinguish high priority users from normal network users.

5.1.4.2 Integrity (AP4-I)

3GPP Rel-15 as the first 5G specification release mandates the user equipment to support integrity protection on the user plane for data rates up to 64 kbps [2]. This limitation was introduced as a compromise to limit battery drain on the user devices at higher data rates but enables an attacker to tamper data if the data rate exceeds 64 kbps. 3GPP Rel-16 removes this limitation and mandates the user equipment to support integrity protection on the user plane up to the maximum data rate supported by the device. It is unclear, however, whether or when the manufacturers of device modem chipsets will support integrity protection up to the maximum data rate.

A recent study investigating the security features enabled in real deployments of 5G networks in Europe revealed that, while confidentiality protection is enabled in most of the studied deployments, integrity protection was completely missing [52]. However, the deployments investigated in this study were confined to non-standalone 5G networks in which the security features of the 4G system apply.

5.1.4.3 Availability (AP4-A)

New features of the 5G air interface may enable new opportunities for denial-of-service (DoS) attacks, e.g. by flooding the control plane signaling by exploiting the newly introduced RRC_INACTIVE state to create a massive amount of RRC state transitions, or by exploiting the ability to request system information on demand [6].

Unencrypted downlink information on MIB/SIB on 5G air interface enables to create a fake presence of a 5G base station. This can cause availability issues as the UE attempts to connect to the fake 5G base station but, as network authentication is expected to fail, will not accept service by the fake 5G base station.

5.1.5 Failure of Power Supply (AP5)

Failure of power supply was reported to cause outages of mobile communications networks [70], data centers [31][32][33] and the internet [72]. Communication network providers appear to become more wary about the availability of power supply in the wake of global climate change and geopolitical shifts [34][69].

Failure of power supply may occur on a local, regional, or multi-regional level, and may be caused by human error, power system failure, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. In the wake of global climate change, natural disasters are expected to become more frequent and more severe [59].

5.1.5.1 Confidentiality (AP5-C)

A power failure can cause some data loss, e.g. if the power failure occurs while data is being written to storage like writing to a database.

5.1.5.2 Integrity (AP5-I)

Both over- and undervoltage can cause damage to hardware parts and/or lead to computation errors which may influence data integrity.

5.1.5.3 Availability (AP5-A)

Radio sites of public mobile communication networks are typically equipped with Uninterruptible Power Supply (UPS) such as backup batteries or diesel generators. In Europe, the average UPS capacity per radio site is dimensioned to cover around 30 minutes of power supply failure from the grid [34]. There have been discussions ongoing between the network operators, emergency services providers and governments to enlarge the UPS capacities to power the radio sites for up to two hours, but there are concerns due to the cost implications, particularly for maintenance [34]. UPS equipment deployed at cell sites is also reported an easy target for vandalism or theft [34]. If power supply from the grid fails over a period longer than that covered by the UPS, this will inevitably cause outage of the affected radio site.

Outage of one or a small number of cell sites will degrade network performance, as the users need to be served with a reduced number of cells, and some users will lose their preferred cell and will be served by a less preferred cell. Outage of one or a small number of cell sites may lead to service outage for users having coverage to one or a few cell sites only, users in sparsely populated areas or indoor users, in particular.

Regional or multi-regional failure of power supply may cause outage of the communications network by disabling the cell sites over a wider area. Outage of the communications network over a wider area may also occur due to failure of power supply at a data center hosting RAN elements further away from the cell sites, e.g., the Centralized Unit (CU) of the 5G RAN may be hosted in a regional data center [1].

5.1.6 Failure of Hardware Components or Unavailability of Spare Parts (AP6)

Failure of hardware components was reported the single largest root cause of telecom security incidents in the European Union in the year 2021 [43]. The failure of the VISA payment system may serve as a prominent example of a hardware failure causing damage to the users of the system [35].

Due to the complexity of the RAN there is a wide variety of hardware components that may cause a failure, such as the failure of the remote (RU), distributed (DU), or centralized units (CU) or their components, or failures of routing or switching equipment, power surges, cable breaks, physical damage of connectors, masts, etc. While hardware failures may occur randomly, there are additional failure sources such as extended hardware lifecycles, human error, lightning strikes or natural disasters, terror attacks, e.g. [40], or sabotage or theft of components, e.g. [41]. In the wake of global climate change, natural disasters are expected to become more frequent and more severe [59].

Unavailability of spare parts may occur if supply chains experience disruptions, particularly if supply chains are not well diversified [38]. From a European perspective we can identify the following supply chain risks for the RAN:

- The supply chain for high-end processor chips for data centers and smartphones, is characterized by a concentration of the manufacturing in Taiwan [16]. The respective product offerings by European semiconductor vendors as well as the respective manufacturing capacities in Europe are limited [17][18]. The following statement from [17] may be used to highlight the issue: “There is no European mobile system-on-a-chip of the type used in smartphones; no EU AI accelerator (the bit of the chip for machine learning) with substantial market share; and no European general purpose processor, graphic chip or data center processor.” Concentration of manufacturing may also apply to other specialized components such as radio frequency or sensor chips.
- Financial risk may jeopardize the supply of networking gear by European mobile network equipment vendors [19][20].

Supply chain disruptions may be due to component shortages, interrupted or impaired shipping routes, or other production or distribution issues. Examples include component shortages caused by natural disaster such as an earthquake impairing semiconductor manufacturing in Taiwan [45], the destruction of Baltimore’s Key Bridge involving technical failure [44], the blockage of the Suez channel by a container ship

likely involving human error or technical failure [30], geopolitical tensions affecting the Red Sea trade route [37], or the supply chain upheaval during the COVID-19 pandemic caused by labour shortages, unscheduled closure of production and distribution sites, irregular demand, and unfavourable distribution of shipping containers [46].

Loss of shipping containers at sea is reported as having minor impact on supply chain disruptions, with a yearly average of 2301 containers lost over the years 2020-2022 versus 250 million containers transported in 2022 [47]. The number of containers lost per year fluctuates significantly as this number correlates with the occurrence of single accidental events such as losses of entire container vessels [47], and reached a high during the COVID-19 pandemic with close to 4000 containers lost in 2020 [47][46][48]. The loss of shipping containers at sea can have devastating impact on ocean habitats and marine life, though [48].

5.1.6.1 Confidentiality (AP6-C)

Failure of hardware components / unavailability of spare parts is not expected to cause confidentiality issues with 5G RAN.

5.1.6.2 Integrity (AP6-I)

Failure of hardware components / unavailability of spare parts is not expected to cause integrity issues with 5G RAN.

5.1.6.3 Availability (AP6-A)

Failure of hardware components / unavailability of spare parts may cause impairment or outage of one or multiple radio cells or radio cell sites, e.g., failure of a radio transceiver may disable one or multiple antennas, failure of a baseband card may disable one or multiple cells, or failure of a network termination card may disable one or multiple cells or cell sites. This will generally lead to degraded network performance. Outage of one or a small number of cells may also lead to service outage for users having coverage to one or a few cells only, users in sparsely populated areas or indoor users, in particular. A hardware failure causing regional network outage would need to occur at a more centralized location in the network, away from the cell sites, for example, a server hosting a cloudified Centralized Unit (CU) or an aggregation router or switch located in a regional data center.

5.1.7 Radio Interference (AP7)

Any radio network has the inherent weakness that it can be attacked by jamming the air interface with electromagnetic interference signals. The cost and complexity of jamming one or a small number of radio cells or sites can be considered as low to medium, while jamming a radio network over a wider geographical area would require a substantial amount of resources and capabilities. Jamming attacks may impair network performance or result in outage of one or several radio cells, depending on the scale of the attack. Jamming attacks are commonly applied as part of electronic warfare [36].

Jamming attacks can be carried out with commercial or proprietary jamming devices, i.e., radio transmitters specifically designed for jamming purposes. While their legal use is restricted to exceptional scenarios, jamming devices can be acquired with low cost and effort. Jamming devices may exhibit different degrees of sophistication, e.g., jam a certain part of the frequency spectrum (wideband jamming) or be tailored to jam certain radio signals such as the initial access signals of the 5G NR air interface (narrowband jamming).

Besides the air interface of the mobile communications system itself, also auxiliary GNSS receivers deployed in a network can be targets of jamming attacks. Mobile communications systems often apply GNSS receivers for time synchronization purposes, particularly if using time division duplex in an unpaired band such as the widely used C-Band [49]. In Germany, for example, the unpaired band n78 (3300 – 3800 MHz) is used for both public and private 5G network deployments in the C-Band. In a public network, a GNSS

receiver may be deployed per outdoor cell site, while for a private campus network, a GNSS receiver may provide time synchronization for the entire network. Alternative synchronization sources like timing over packet may be used in a network as an alternative or complement to GNSS-based synchronization, particularly in public networks. Depending on the implementation of clock synchronization and the scale of the attack, jamming of GNSS receivers may impair performance or lead to outage of one or several cell sites of a radio access network, or of the entire network in case of a private campus network.

Flooding the air interface of the mobile communication system with data packets transmitted from a potentially large number of compromised devices, e.g., smart phones or Internet of Things (IoT) devices, is a type of botnet attack utilizing radio interference [7][8]. In this case, the compromised (or “weaponized”) devices may generate large traffic volumes which, depending on the number and the geographical distribution of the involved devices, may lead to congestion or outage of parts of the access network.

Excessive radio interference may also be caused by poor network design or flawed network configuration, but such scenarios are considered as implementation flaws or misconfigurations, respectively.

5.2 Public Mobile Network Operator (MNO) Perspective

This section discusses the risk scenarios related to the MNOs perspective.

For each of the following subsections, a short evaluation of the relevance of the risk scenarios in the RAN domain is discussed and the probability that potential attack paths could be exploited is derived.

5.2.1 Emergency Call (TKG § 164)

An attacker could attack the RAN in a way leading to unavailability of one or some base stations. In the case of a physical attack, even all radio technologies of that MNO could be impacted. It is likely that also other MNOs use some of the assets (e.g. tower poles, cabinets) together with the attacked MNO, so that the service of the other MNOs may also be affected. But it is also rather likely that most geographical areas are still covered by network elements of other mobile network technologies or neighboring sites. Furthermore, even if a particular site is being attacked and the RAN of all MNOs would be unavailable at this site, there is a high probability that at least one MNO's network can still provide service by any surrounding network element to the user, because MNOs normally use different network structures / frequencies.

So, **RAN is not considered** a relevant attack domain for emergency call services.

5.2.2 Cell Broadcast (TKG § 164a)

An attacker could attack the RAN in a way leading to unavailability of one or several base stations. One or multiple or all radio technologies of the MNO could be affected. In the latter case, there could be geographical areas, indoor locations in particular, where reception of cell broadcast messages over any radio technology is not possible.

As cell broadcast services are normally intended to be transmitted to many or even all persons in a particular geographical area and the messages are usually of high importance, the **RAN is considered** a relevant attack domain for cell broadcast services.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Cell broadcast messages are transmitted via System Information Broadcast (SIB6, SIB7, SIB8 messages). RAN misconfiguration may affect the mapping of the message payload to system information (SI), or the configuration or scheduling parameters of the system information transmission such as SI periodicity, and it may lead to loss of SI messages [85]. As dedicated SI configuration parameters are few in numbers and can be assumed as correctly configured per default, the probability of misconfiguration is assumed as LOW. There are many other configurable radio parameters, but those are largely shared with other SIB messages or downlink data, and any such misconfiguration would likely be detected during normal operations. Misconfiguration may also affect the interworking between the core network and the RAN, particularly with multi-vendor deployments, or the respective transport links, e.g. addresses and/or port numbers. Such misconfigurations are attributed to the core or transport domains, respectively.	Low
Insufficient Access Controls	Availability (AP2-A)	Adversaries may disable base stations which may impair or disable cell broadcast in the affected area, e.g. in conjunction with a terror attack. Given that such a coordinated attack would require special resources, skills, and sophistication, it is rated as having LOW probability. Public warning messages related to natural disasters appear unrelated with access breaches.	Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered having a MEDIUM probability to impair or disable cell broadcast, as any flaw may remain undetected due to the rare occurrence of public warning messages. Such flaws may have a variety of causes, for example: • SW upgrades can lead to the use of flawed or misconfigured SW [24][25]. • Interoperability issue in multi-vendor deployment, e.g. core network and RAN may be provided by different vendors. Cell broadcast was tested in Germany in the year 2022 for the first time during the yearly public warning test event, with 90% of the population successfully receiving the warning, and 54% of the population receiving the warning via cell broadcast messages [42].	Medium
Specification Weakness	Integrity (AP4-I)	Master and System Information Blocks (MIB/SIB) are neither encrypted nor integrity protected which in principle enables attacks on the cell broadcast, e.g. by man-in-the-middle attacks [39]. Due to the high effort required to compromise cell broadcast over a larger area via a RAN attack exploiting such specification weaknesses, this is rated as having LOW probability.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	Natural disasters such as flooding, fire or earthquakes are relevant causes of public warning messages being transmitted via cell broadcast. Such disaster events can simultaneously cause power outages, possibly over larger geographical areas. In such cases, all base stations in the affected areas are cut off from the power supply and can't transmit cell broadcast messages. So the probability that cell broadcast is impaired or disabled by a power supply failure is rated as HIGH .	High
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Natural disasters such as flooding, fire or earthquakes are relevant causes of public warning messages being transmitted via cell broadcast. Such disaster events can simultaneously cause hardware failures, possibly over larger geographical areas. However, in such cases, it is unlikely that fatal hardware failures occur at all base stations in the affected areas, so that public warning messages can no longer be sent. So, in contrast to failure of power supply, the probability that cell broadcast is impaired or completely disabled by a hardware failure is rated as MEDIUM .	Medium
Radio Interference	Radio Interference (AP7)	Adversaries may disable or interfere the radio transmission of one or several base stations which may impair or disable cell broadcast services, e.g. in conjunction with a terror attack. Given that such a coordinated attack would require special resources, skills, and sophistication, it is rated having a LOW probability.	Low

5.2.3 Telecommunications Secrecy (TTDSG § 3)

There are weaknesses that may allow an attacker to gain access to user-related data by observing data flows at the air interface, at the network interfaces for backhaul, midhaul, fronthaul, or interhaul, or by intercepting at the base station directly.

So, **RAN is considered** a relevant attack domain for telecommunication secrecy.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	RAN software is often pre-configured with default configurations that can't be assumed to be secure regarding ciphering and integrity protection [88]. Even if encryption and integrity protection are enabled in the RAN by the vendor or the network operator, these parameters may be disabled on the air or network interfaces by misconfiguration. However, to conduct a successful attack on configurations, an adversary has to gain access to the RAN components first. Therefore, the probability of misconfiguration issues regarding ciphering is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Confidentiality (AP2-C)	RAN components that use cloud-based solutions might be compromised due to insufficient access controls. Those components can then be re-configured by an attacker to use insecure RAN configurations or by adding backdoors or weakening cryptographic elements. This is also possible through physical access to non-cloud components. Considering an attacker has direct access to RAN components, they only hold limited user-specific information apart from communication payloads. In general, communication payloads can be considered as protected by measures on both the transport and application layer. For phone calls, the operator is responsible for the encryption and integrity protection of respective signaling and communication payloads as its use is optional for MNOs. Consequently, it can be assumed that a respective encryption is not enabled by default, which could allow an attacker to eavesdrop phone calls. The probability of such attacks is rated as LOW, because attackers need further knowledge about accessing or manipulation of the software on the hardware after accessing the RAN and attacks have to be highly sophisticated and are generally not very beneficial.	Low
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Software updates can affect the configuration parameters and introduce new parameters with corresponding new default values. Updates may also overwrite existing parameters. In both cases, this could lead to incorrect configuration of the RAN components, which means that user data is not adequately protected. Software updates can be assumed to be well tested and changes to existing parameters can be assumed to occur rarely. Access to keys and identities through implementation flaws can allow an attacker to track users. This might not be very beneficial as the identities would need to be correlated with personal data (such as employee-specific information) from outside the RAN to be useful for a user mapping or for further data extractions. Further, such an access can for instance be utilized through supply chain attacks that contain cryptographic backdoors with weakened cryptography. As attacks on opensource software become more common and more sophisticated, vendors have to develop a development and deployment strategy. Because of the reasons mentioned above, the probability of this scenario is rated as MEDIUM.	Medium
Specification Weakness	Confidentiality (AP4-C)	User and control plane can be protected by encryption, which is an optional feature for MNOs. Both, signalling and the payload might therefore not be encrypted. It can be assumed, that encryption is not enabled by default, which could allow an attacker to eavesdrop signalling traffic and communication contents. Therefore, the probability of this scenario is rated as MEDIUM.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Confidentiality (AP5-C)	The failure of power supply appears unrelated to telecommunications secrecy.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	The failure of hardware components or unavailability of spare parts appears unrelated to telecommunications secrecy.	Low
Radio Interference	Radio Interference (AP7)	Attacks utilizing radio interference are not directly related to telecommunications secrecy. However, radio interference can be used for downgrade attacks forcing the UE to connect to other mobile networks which don't provide state-of-the-art security (e.g. 2G networks). Because it is unlikely that radio interference is performed on a large scale over wider geographical areas, the probability that radio interference is utilized to gain access to user-related data is LOW .	Low

5.2.4 Emergency Preparedness (TKG § 184)

For the requirement of emergency preparedness (§ 184 TKG), mobile network operators have to ensure the availability of services by prioritizing predefined users such as police or first responder teams in the event of a disaster.

Because of that, malfunctioning prioritization of predefined users or an unavailability of the **RAN** could have severe impact and is therefore **considered** a relevant attack domain for emergency preparedness.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Emergency preparedness requires load detection for the radio cell and respective barring of the cell for normal users in case of high load. Depending on the network implementation, this mechanism may be automated without intervention by the network operations team. In this case, misconfiguration may affect the cell load threshold. Alternatively, the mechanism may be monitored and configured by the network operations team. In this case, the cell barring may be deactivated by misconfiguration despite of high load. Dedicated configuration parameters related with cell load are few in numbers and can be assumed as correctly configured per default, therefore the probability of misconfiguration is assumed to be LOW. Misconfiguration can lead to degraded cell performance in general. However, such performance degradation can be observed during normal operations.	Low
Insufficient Access Controls	Availability (AP2-A)	Adversaries may disable base stations e.g. in conjunction with a terror or military [71] attack to compromise work of police or first responders. As cyber-attacks, e.g. [50][51], and physical attacks, e.g. [40][41], on technical infrastructure have become more widespread the probability of impaired or disabled emergency preparedness being caused by insufficient access controls is rated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to impair or disable emergency preparedness, as such flaws may have a variety of causes and may be non-trivial to detect, for example: • Radio admission control is based on proprietary algorithms of the network vendor and may not work as expected in emergency situations. Implementation flaws may remain undetected due to the rare occurrence of emergency situations. • SW upgrades can lead to the use of flawed or misconfigured SW [24][25]. • The implementation may be compromised, e.g. compromised open-source SW may be utilized, particularly if using a cloud platform [14][15][74].	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Availability (AP4-A)	There are some specification weaknesses in 5G RAN which may enable attacks leading to impaired or disabled emergency preparedness, for example: - Cell barring is indicated on MIB without encryption or integrity protection, which may be exploited by a man-in-the-middle attack [39]. - DoS attacks exploiting 5G particularities (e.g. by causing frequent RRC state transitions [6]) or using compromised UEs emulating high priority users [9], would require many UEs. Those attacks have in common that the effort to conduct them is high with respect to the achievable gain and impacted geographical area. Therefore, such attacks are rated having a LOW probability.	Low
Failure of Power Supply	Availability (AP5-A)	Natural disasters such as flooding, fire or earthquakes are relevant causes for emergency scenarios. Such disaster events can simultaneously cause power outages, possibly over larger geographical areas. Further, also adversaries may disable power supply, e.g. in conjunction with a terror attack to compromise the work of police or first responder teams. The arson attack on energy infrastructure at Grünheide [40] may serve as an example. However, it can be assumed that the police or civil protection organisations mostly communicate over well-protected, non-public wireless networks which do not rely on the power supply of the public MNO. Therefore, the probability of this scenario is rated as MEDIUM.	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Natural disasters such as flooding, fire or earthquakes are relevant causes for emergency scenarios. Such disaster events can simultaneously cause hardware failures, possibly over larger geographical areas. However, in such cases, it is unlikely that fatal hardware failures occur at all base stations in the affected areas, so that user prioritization can not be performed anymore. Therefore, the probability of this scenario is rated as LOW.	Low
Radio Interference	Radio Interference (AP7)	DoS attacks utilizing radio interference may be carried out by means of jamming or by using compromised devices. The effort to conduct such attacks is high with respect to the achievable gain and impacted geographical area. Therefore, such attacks are rated having a LOW probability.	Low

5.2.5 Regional Network Failure

An attack to hundreds of RAN elements leading to a regional unavailability of the network appears a realistic attack path, particularly if an internal attacker performs a complex attack.

The **RAN** is therefore **considered** a relevant attack domain for a regional network failure.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	A regional network failure may be caused by, for example: • Misconfiguration via 5G RAN management system or central network configuration systems affecting many base stations. • Misconfiguration of cloudified Centralized Unit (CU) hosted in a regional data center. • Misconfiguration of aggregation routers or switches in the RAN transport network. Given that multiple configuration domains with a wide range of parameters and multiple responsible individuals can be involved, the probability of a misconfiguration causing a regional network failure is rated as MEDIUM.	Medium
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause a regional network failure by gaining access to the 5G RAN management system or central network configuration systems, e.g. by means of a cyber-attack, or by physically attacking a regional data center hosting servers used by the telecom network. As cyber-attacks, e.g. [50][51], and physical attacks, e.g. [40][41], on technical infrastructure have become more widespread the probability of a regional network failure being caused by insufficient access controls is rated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered having a MEDIUM probability to cause regional network failures, as such flaws may have a variety of causes and may be non-trivial to detect, for example: • SW upgrades can lead to the use of flawed or misconfigured SW [24][25]. • The implementation may be compromised, e.g. compromised open-source SW may be utilized, particularly if using a cloud platform [14][15][74]. Faulty SW updates were the single largest cause of telecom user hours lost in the EU in the year 2022 [10].	Medium
Specification Weakness	Availability (AP4-A)	There are some specification weaknesses which may enable attacks on the 5G RAN, for example: • Cell barring is indicated on the Master Information Block (MIB) without being encrypted or integrity protected, which may be exploited by a man-in-the-middle attack [39]. • DoS attacks exploiting 5G particularities (e.g. causing frequent RRC state transitions [6]) or using compromised UEs emulating high priority users [9], would require many UEs. Causing a regional network failure by the DoS attacks described above would require high effort and is therefore considered having a LOW probability.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Availability (AP5-A)	Regional network failure may be caused by regional power outage, or by power outage in a data center [31][32][33][34]. A regional power outage may be caused by human error, power system failure, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Because failure of power supply is a frequently observed cause for network outage [10] and given the variety of different sources for failures of power supply, the probability to be the cause of a regional network failure is rated as HIGH .	High
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	A hardware failure causing regional network outage would need to occur at a more centralized location in the network, away from the cell sites, for example, a server hosting a cloudified Centralized Unit (CU) or an aggregation router or switch located in a regional data center. Even if data center processing hardware and routed/switched transport links provide redundant hardware resources and transmission routes, respectively, failure of hardware components is a frequently observed cause for network outage [10]. Therefore, the probability of a hardware failure causing a regional network failure is rated as MEDIUM .	Medium
Radio Interference	Radio Interference (AP7)	To carry out DoS attacks by means of radio interference over a wider geographical area is considered high effort and therefore rated as having LOW probability to cause a regional network failure.	Low

5.2.6 Multi-Regional Network Failure

An attacker could use the RAN as a gateway to the core network functions to perform denial-of-service attacks in the core network with the aim to impair or disable the network over a large geographical area. There are potential vulnerabilities that could enable such an attack. But as the core network functions in a public network are dimensioned to serve huge amounts of traffic/requests, it is very unlikely that an attacker could successfully perform such a distributed denial-of-service attack, that really could break down relevant core network functions.

The **RAN** is therefore **not considered** a relevant attack domain for multi-regional network failure.

5.2.7 Failure of Mobile Service

In general, mobile services are hosted in the 5G core network. The RAN is only responsible for radio admission and radio resource control for different mobile services. However, this scenario describes a situation where significant parts or even the entire 5G network is impacted by a failure of mobile services. Attacks to the RAN or malfunctions of the RAN only affect the attacked base stations. As attacks to thousands of base stations originating in the RAN appear impossible, the **RAN** is therefore **not considered** a relevant attack domain for failure of mobile services.

5.2.8 Fraud

Any billing related information, subscription data and (temporary) user identities are processed and/or assigned by the core network.

So, by attacking the RAN it appears impossible to perform a fraud attack.

The **RAN** is therefore **not considered** a relevant attack domain for fraud.

5.2.9 Subscriber-Data Leak

Data leak is defined as the leakage of subscriber-related data from a significant number of users, which could occur if an attacker could get access to confidential data of users of many base stations.

The **RAN** is therefore **considered** a relevant attack domain for data leak.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	Encryption and integrity protection may be disabled on the air or network interfaces by misconfiguration. RAN software is often pre-configured and contains default configurations that can't be assumed to be secure regarding ciphering and integrity protection [88]. For internal attacks, to conduct a successful attack on configurations, an adversary has to gain access to the RAN components. As the number of configurable parameters related to subscriber-related data is limited and it takes a lot of effort for an attacker to carry out internal attacks, the probability of misconfiguration issues is therefore rated as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Confidentiality (AP2-C)	RAN components that use cloud-based solutions might be compromised due to insufficient access controls. Those components can then be re-configured by an attacker to use insecure RAN configurations or by adding backdoors or weakening cryptographic elements. This is also possible through physical access to non-cloud components. Considering an attacker has direct access to RAN components, they only hold limited subscriber-related data apart from communication payloads. In general, communication payloads can be considered as protected by measures on both the transport and application layer. The probability of such attacks is rated as LOW, because attackers need further knowledge about accessing or manipulation of the software on the hardware after accessing the RAN and attacks have to be highly sophisticated and are generally not very beneficial.	Low
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Software updates can affect the configuration parameters and introduce new parameters with corresponding new default values. Updates may also overwrite existing parameters. In both cases, this could lead to incorrect configuration of the RAN components, which means that user data is not adequately protected. Software updates can be assumed to be well tested and changes to existing parameters can be assumed to occur rarely. Access to keys and identities through implementation flaws can allow an attacker to track users. This might not be very beneficial as the identities would need to be correlated with personal data (such as employee-specific information) from outside the RAN to be useful for a user mapping or for further data extractions. Further, such an access can for instance be utilized through supply chain attacks that contain cryptographic backdoors with weakened cryptography. As attacks on opensource software become more common and more sophisticated, vendors have to develop a development and deployment strategy. For these reasons, the probability of this scenario is rated as MEDIUM.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Confidentiality (AP4-C)	User priority is transmitted on the air interface without encryption, which allows attackers to track groups of priority users [9], but not individuals specifically, which in most situations provides little benefit to attackers. Signaling of phone calls can be protected by encryption, which is an optional feature for MNOs. Both, signaling and the payload might therefore not be encrypted. It can be assumed, that encryption is not enabled by default, which could allow an attacker to eavesdrop calls. The probability is rated as LOW, because attackers can track priority users (e.g. police movements) with medium effort and downgrade attacks are only possible potentially and under certain circumstances [3].	Low
Failure of Power Supply	Confidentiality (AP5-C)	The failure of power supply appears unrelated to leakage of subscriber-related data.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	Hardware failures appear unrelated to leakage of subscriber-related data.	Low
Radio Interference	Radio Interference (AP7)	DoS attacks by means of radio interference appear unrelated to leakage of subscriber-related data.	Low

5.3 User Perspective

This section describes different risk scenarios from the perspective of a user of the public mobile network. Private 5G networks are not in scope of this section. Since some of the risk scenarios one might typically expect here (e.g. emergency calls) are described in the previous paragraphs, they will not be covered in this section.

5.3.1 Urgent Telephone Calls Not Possible

The risk scenario of urgent telephone calls not possible may be related with a regional or other type of network failure, e.g. a localized network failure may disable users to make an urgent phone call. An attack to one or several RAN network elements appears a realistic attack path.

So, the **RAN is considered** a relevant attack domain for the risk scenario of urgent telephone calls not possible.

Note: Urgent telephone calls may be carried out by using voice telephony over the 5G System based on an IMS core, often called VoNR. VoNR got standardized by 3GPP starting with Rel-15 [60]. First VoNR deployments got announced by MNOs around mid-2023 [61][62][63]. Both application layer and transport layer encryption for VoNR are assumed to be an optional feature that can be enabled by the network operator, but due to limited information publicly disclosed on the topic, it is assumed as being typically disabled.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Given that multiple configuration domains including the central RAN configuration management, the cloudified Centralized Unit (CU) or the RAN transport network elements with a wide range of configuration parameters and multiple responsible individuals can be involved, the probability of a misconfiguration causing urgent telephone calls not possible is rated as MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause a regional network failure by gaining access to the central network configuration management system, e.g. by means of a cyber-attack, or by physically attacking a regional data center hosting servers used by the telecom network. Further, an adversary may cause a localized network failure by physical access to a configuration interface or by physically attacking distributed network sites. As cyber-attacks, e.g. [50][51], and physical attacks, e.g. [40][41], on technical infrastructure have become more widespread the probability of urgent telephone calls not possible resulting from a regional or local network failure being caused by insufficient access controls is rated as MEDIUM .	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered having a MEDIUM probability to cause urgent telephone calls not possible, as such flaws may have a variety of causes and may be non-trivial to detect, for example: - SW upgrades can lead to the use of flawed or misconfigured SW [24][25]. - The implementation may be compromised, e.g. compromised open-source SW may be utilized, particularly if using a cloud platform [14][15][74]. Faulty SW updates were the single largest cause of telecom user hours lost in the EU in the year 2022 [10].	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Availability (AP4-A)	There are some specification weaknesses which may enable DoS attacks on the 5G RAN, for example: - Cell barring is indicated on the Master Information Block (MIB) without being encrypted or integrity protected, which may be exploited by a man-in-the-middle attack [39]. - DoS attacks exploiting 5G particularities (e.g. causing frequent RRC state transitions [6]) or using compromised UEs emulating high priority users [9], would require many UEs. Causing a regional or localized network failure by the DoS attacks above would require high effort and is therefore considered having a LOW probability to result in urgent telephone calls not possible.	Low
Failure of Power Supply	Availability (AP5-A)	Regional or localized network failure may be caused by regional or localized power outage, or by power outage in a data center [31][32][33][34]. Power outage may be caused by human error, power system failure, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Given the variety of different sources for failures of power supply, their probability to be the cause of urgent telephone calls not possible is rated as MEDIUM.	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to cause a localized network failure leading to urgent telephone calls not possible is rated as MEDIUM, as there are a wide variety of components that may cause the failure, such as failed Remote (RU), Distributed (DU) or Centralized Units (CU), routing or switching equipment, cable breaks, physical damage of connectors, masts, etc., and a variety of failure sources such as human error, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. HW failures were the single largest root cause of telecom security incidents in the EU in the year 2021 [43].	Medium
Radio Interference	Radio Interference (AP7)	Voice telephony has low data rate requirements (e.g. 5.2 to 48 kbps with the recent Extended Adaptive Multi-Rate – Wideband AMR-WB+ audio codec [87]) and is therefore generally robust against the presence of radio interference. Any DoS attack utilizing radio interference, e.g. by means of jamming or by deploying compromised devices, would therefore require to entirely disable one or more radio cells to cause urgent telephone calls not possible. The effort to conduct such attacks is high with respect to the achievable gain and impacted geographical area. Therefore, such attacks are rated having a LOW probability.	Low

5.3.2 Authenticity of the Receiver of a Call

The RAN is not involved into the routing of calls. So, by attacking the RAN it appears impossible to route a call to non-intended receivers.

The **RAN** is therefore **not considered** an attack domain for undermining the authenticity of the receiver of a call.

5.3.3 Integrity of Transmitted Data

There are vulnerabilities within the RAN that could lead to user data not being integrity-protected.

The **RAN** is therefore **considered** a relevant attack domain for integrity of transmitted data.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Integrity (AP1-I)	RAN software is often pre-configured with default configurations that can't be assumed to be secure regarding ciphering and integrity protection [88]. Even if encryption and integrity protection are enabled in the RAN by the vendor or the network operator, these parameters may be disabled on the air or network interfaces by misconfiguration. However, to conduct a successful attack on configurations, an adversary has to gain access to the RAN components first. Therefore, the probability of misconfiguration issues regarding integrity protection is rated as MEDIUM .	Medium
Insufficient Access Controls	Integrity (AP2-I)	RAN components that use cloud-based solutions might be compromised due to insufficient access controls. Those components can then be re-configured by an attacker to use insecure RAN configurations. This is also possible through physical access to non-cloud components. A main difference between physical and virtual access is, that after accessing the RAN components physically, further action and know-how is needed to access and manipulate the software running on the hardware. RAN components hold only limited user-specific information apart from communication payloads. In general, communication payloads can be considered as protected by measures on both the application layer and transport layer. For phone calls, the operator is responsible for the encryption and integrity protection of respective signaling and communication payloads as its use is optional for MNOs. Consequently, it can be assumed that a respective integrity protection is not enabled by default, which could allow an attacker to violate the integrity of phone calls. Attackers need further knowledge about accessing or manipulation of the software on the hardware after accessing the RAN attacks have to be highly sophisticated and are generally not very beneficial. The probability of attacks is therefore rated as LOW.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Integrity (AP3-I)	An attacker can include backdoors on cryptographic algorithms, keys or procedures via supply chain attacks. This could be used to target the integrity of transmitted data. Supply chain attacks are becoming more common and more sophisticated. In contrast, transmitted data are often additionally encrypted and authenticated at the application layer. For communication data of phone calls, the protection of the integrity is optional and can be considered as not enabled by default. As attacks require a high level of sophistication and effort the probability of this scenario is rated as LOW.	Low
Specification Weakness	Integrity (AP4-I)	Between UE and RAN, data rates above 64kbps are not integrity protected in Release 15. As MNOs are untransparent regarding their implementations, Release 15 implementations can be considered to be still in use. Further it has to be considered, that UEs might not be able to support the integrity protection for data rates above 64kbps yet. For high data rates it depends on the specific implementation, whether the user data is protected or not. Transmitted data is often additionally encrypted and authenticated at the application layer. For communication data of phone calls, the protection of the integrity is optional and can be considered as not enabled by default. In this case, as phone calls have low data rate requirements, it can be considered that data rates of phone calls are below 64kbps [87]. For these reasons attacks are not very beneficial. The probability of this scenario is therefore rated as LOW.	Low
Failure of Power Supply	Integrity (AP5-I)	The failure of power supply appears unrelated to the integrity protection of transmitted data.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Integrity (AP6-I)	The failure of hardware components or the unavailability of spare parts appears unrelated to the integrity protection of transmitted data.	Low
Radio Interference	Radio Interference (AP7)	Radio interference appears unrelated to the integrity protection of transmitted data.	Low

5.3.4 User's Identity

There are multiple attack vectors on the user's identity. Misconfiguration issues could lead to revealing the SUPI. But as the user's identity is not solely tied to the SUPI, also other information like communication payloads or metadata can reveal information about the user's identity. Specific user groups can be identified

more easily than standard users due to specification weaknesses. Additionally, other specification weaknesses can allow to eavesdrop the SUPI under certain circumstances.

So, the **RAN** can be **considered** a relevant attack domain for revealing and/or abusing the user's identity.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	Encryption and integrity protection between the RAN and the UE might be disabled on the air or network interfaces by misconfiguration. While the SUPI as a part of the user's identity is protected by a concealed identifier, the SUCI, unprotected data transmissions can be utilized to extract information about the user. Furthermore, if the SUCI feature is disabled by a misconfiguration, the SUPI can be extracted directly. The extraction of information on the SUPI requires highly sophisticated attacks with high effort and certain circumstances. Therefore, the probability of misconfiguration issues that can reveal a user's identity is rated as LOW .	Low
Insufficient Access Controls	Confidentiality (AP2-C)	Insufficient access control to RAN components, their software or shared resources in the RAN could enable an attacker to access sensitive information that might be helpful to extract the user's identity. As RAN components hold only limited user-specific information the probability is rated as LOW .	Low
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	An attacker can include backdoors on cryptographic algorithms, keys or procedures via supply chain attacks. This could be used to weaken the cryptography which could then allow attackers to extract information about the SUPI or the user. Further, infrequent updates of temporary user identifiers might enable attackers to track a user's position [11]. As these attacks require high effort and provide low benefit, the probability is therefore rated as LOW .	Low
Specification Weakness	Confidentiality (AP4-C)	User priority is transmitted on the air interface without encryption, which allows attackers to track groups of priority users [9], but not individuals specifically, which in most situations provides little benefit to attackers. Further, unprotected data before authentication could be used to fingerprint UEs. The probability is rated as LOW , because attackers can track priority users (e.g. police movements) with medium effort and downgrade attacks are only possible potentially and under certain circumstances [3].	Low
Failure of Power Supply	Confidentiality (AP5-C)	The failure of power supply appears unrelated to the exposure of a user's identity.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	The failure of hardware components or unavailability of spare parts appears unrelated to the exposure of a user's identity.	Low
Radio Interference	Radio Interference (AP7)	Radio interference appears unrelated to the exposure of a user's identity.	Low

5.3.5 Eavesdropping of a Communication

There are potential implementation flaws and specification weaknesses that could lead to the deactivation of encryption or to the use of weak encryption at the air or network interfaces.

So, the **RAN is considered** a relevant attack domain for the risk scenario of eavesdropping of a communication.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	Encryption and integrity protection may be disabled on the air or network interfaces by misconfiguration. RAN software is often pre-configured and contains default configurations. It can be assumed that ciphering on the user plane is activated in mobile networks of european MNOs [88]. However, even if encryption and integrity protection are enabled in the RAN by the vendor or the network operator, these parameters may be disabled on the air or network interfaces by misconfiguration. But, to conduct a successful attack on configurations, an adversary has to gain access to the RAN components first. Therefore, the probability of misconfiguration issues regarding ciphering on the user plane is rated as LOW .	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Confidentiality (AP2-C)	RAN components that use cloud-based solutions might be compromised due to insufficient access controls. Those components can then be re-configured by an attacker to use insecure RAN configurations or by adding backdoors or weakening cryptographic elements. This is also possible through physical access to non-cloud components. Considering an attacker has direct access to RAN components, they only hold limited user-specific information apart from communication payloads. In general, communication payloads can be considered as protected by measures on both the transport and application layer. For phone calls, the operator is responsible for the encryption and integrity protection of respective signaling and communication payloads as its use is optional for MNOs. Consequently, it can be assumed that a respective encryption is not enabled by default, which could allow an attacker to eavesdrop phone calls. The probability of such attacks is rated as LOW, because attackers need further knowledge about accessing or manipulation of the software on the hardware after accessing the RAN and attacks have to be highly sophisticated and are generally not very beneficial.	Low
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Software updates can affect the configuration parameters and introduce new parameters with corresponding new default values. Updates may also overwrite existing parameters. In both cases, this could lead to incorrect configuration of the RAN components, which means that user data is not adequately protected. Software updates can be assumed to be well tested and changes to existing parameters can be assumed to occur rarely. Access to keys and identities through implementation flaws can allow an attacker to track users. This might not be very beneficial as the identities would need to be correlated with personal data (such as employee-specific information) from outside the RAN to be useful for a user mapping or for further data extractions. Further, such an access can for instance be utilized through supply chain attacks that contain cryptographic backdoors with weakened cryptography. As attacks on opensource software become more common and more sophisticated, vendors have to develop a development and deployment strategy. For these reasons, the probability of this scenario is rated as MEDIUM.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Specification Weakness	Confidentiality (AP4-C)	User priority is transmitted on the air interface without encryption, which allows attackers to track groups of priority users [9], but not individuals specifically, which in most situations provides little benefit to attackers. Signaling of phone calls can be protected by encryption, which is an optional feature for MNOs. Both, signaling and the payload might therefore not be encrypted. It can be assumed, that encryption is not enabled by default, which could allow an attacker to eavesdrop calls. The probability is rated as LOW, because attackers can track priority users (e.g. police movements) with medium effort, but downgrade attacks are only possible potentially and under certain circumstances [3] and the eavesdropping of phone calls requires access to the communication payload.	Low
Failure of Power Supply	Confidentiality (AP5-C)	The failure of power supply appears unrelated to eavesdropping of communication.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	The failure of hardware components or unavailability of spare parts appears unrelated to eavesdropping of communication.	Low
Radio Interference	Radio Interference (AP7)	Radio interference appears unrelated to eavesdropping of communication.	Low

5.3.6 Slowing Down/Restricting the Connection

An attack to one or several RAN network elements may impair network performance and slow down connections. The risk scenario of slowing down/restricting the connection may be related with a localized or regional degradation of network performance. Performance degradation for a user may also be caused by a failure of one or multiple cells in the network, e.g. the best serving cell(s) for the user could fail, or in case of failed cell(s) the users of the network will be served by a reduced number of active cells.

So, the **RAN is considered** a relevant attack domain for the risk scenario of slowing down/restricting the connection.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Given the complexity and wide configurability of the 5G air interface and the fact that multiple configuration domains including central RAN configuration management, cloudified Centralized Unit (CU) or RAN transport network elements with a wide range of parameters and multiple responsible individuals can be involved, the probability of slowing down or restricting the connection caused by a misconfiguration is rated as MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause impaired network performance by gaining access to the central network configuration management system, e.g. by means of a cyber-attack, or by physically attacking a regional data center hosting servers used by the telecom network. Further, an adversary may cause a localized network performance degradation by physical access to a configuration interface or by physically attacking distributed network sites. As cyber-attacks, e.g. [50][51], and physical attacks, e.g. [40][41], on technical infrastructure have become more widespread the probability of slowing down or restricting the connection resulting from a regional or local network impairment being caused by insufficient access controls is rated as MEDIUM .	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered having a MEDIUM probability to cause restricted connections, as such performance degrading flaws may have a variety of causes and may be non-trivial to detect, for example: • SW upgrades can lead to the use of flawed or misconfigured SW [24][25]. • The implementation may be compromised, e.g. compromised open-source SW may be utilized, particularly if using a cloud platform [14][15][74][74]. • There are many proprietary algorithms impacting network performance, e.g. scheduling, link adaptation, radio admission control, selection of multi-antenna transmission scheme, etc. Network performance also depends on the proper joint operation of these algorithms, which may be flawed e.g. if building new features on top of legacy algorithms.	Medium
Specification Weakness	Availability (AP4-A)	There are some specification weaknesses which may enable performance degradation attacks on the 5G RAN, attacks exploiting 5G particularities (e.g. causing frequent RRC state transitions [6]) or using compromised UEs emulating high priority users [9], in particular. Such attacks would require high effort and sophistication, and they are therefore considered having a LOW probability to result in slowing down or restricting the connection.	Low
Failure of Power Supply	Availability (AP5-A)	Regional or localized network failure may be caused by regional or localized power outage, or by power outage in a data center [31][32][33][34]. Power outage may be caused by human error, power system failure, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Given the variety of different sources for failures of power supply, their probability to be the cause of slowing down or restricting the connection is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to cause a localized network failure that restricts the connection is rated as MEDIUM , as there are a wide variety of components that may cause the failure, such as failed Remote (RU), Distributed (DU) or Centralized Units (CU), routing or switching equipment, cable breaks, physical damage of connectors, masts, etc., and a variety of failure sources such as human error, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Hardware failures were the single largest root cause of telecom security incidents in the EU in the year 2021 [43].	Medium
Radio Interference	Radio Interference (AP7)	A DoS attack utilizing radio interference, e.g. by means of jamming or by using compromised devices, may deteriorate the radio signal quality and therefore lead to restricted connections. The effort to conduct such attacks is high with respect to the achievable gain and impacted geographical area. Therefore, such attacks are rated having a LOW probability.	Low

5.3.7 Connection Failure

Connection failure is considered as the loss of a data connection, e.g. internet access or over-the-top services like instant messaging, voice, or video calls.

The risk scenario of connection failure may be related with a localized or regional network failure, e.g. a localized network failure may disable users to access the internet. Similarly, to chapter 5.3.1 (Urgent Telephone Calls Not Possible) there could be misconfigurations at the RAN (unintended or caused by an attack) that lead to dropping calls or data connections.

So, the **RAN is considered** a relevant attack domain for the risk scenario of connection failure.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Given that multiple configuration domains including the central RAN configuration management, cloudified Centralized Unit (CU) or RAN transport network elements with a wide range of parameters and multiple responsible individuals can be involved, the probability of a misconfiguration causing connection failure is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Availability (AP2-A)	An adversary may cause a regional network failure by gaining access to the 5G RAN management system or central network configuration systems, e.g. by means of a cyber-attack, or by physically attacking a regional data center hosting servers used by the telecom network. Further, an adversary may cause a localized network failure by physical access to a configuration interface or by physically attacking distributed network sites. As cyber-attacks, e.g. [50][51], and physical attacks, e.g. [40][41], on technical infrastructure have become more widespread the probability of connection failure resulting from a regional or local network failure being caused by insufficient access controls is rated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered having a MEDIUM probability to cause connection failures, as such flaws may have a variety of causes and may be non-trivial to detect, for example: - SW upgrades can lead to the use of flawed or misconfigured SW [24][25]. - The implementation may be compromised, e.g. compromised open-source SW may be used, particularly if using a cloud platform [14][15][74][74]. Faulty SW updates were the single largest cause of telecom user hours lost in the EU in the year 2022 [10].	Medium
Specification Weakness	Availability (AP4-A)	There are some specification weaknesses which may enable DoS attacks on the 5G RAN, for example: - Cell barring is indicated on the Master Information Block (MIB) without being encrypted or integrity protected, which may be exploited by a man-in-the-middle attack [39]. - DoS attacks exploiting 5G particularities (e.g. causing frequent RRC state transitions [6]) or using compromised UEs emulating high priority users [9], would require many UEs. Causing a regional or localized network failure by the DoS attacks above would require high effort and is therefore considered having a LOW probability to result in connection failures.	Low
Failure of Power Supply	Availability (AP5-A)	Regional or localized network failure may be caused by regional or localized power outage, or by power outage in a data center [31][32][33][34]. Power outage may be caused by human error, power system failure, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Given the variety of different sources for failures of power supply, their probability to cause connection failure is rated as MEDIUM.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to cause a localized network failure leading to connection failure is rated as MEDIUM, as there are a wide variety of components that may cause the failure, such as failed Remote (RU), Distributed (DU) or Centralized Units (CU), routing or switching equipment, cable breaks, physical damage of connectors, masts, etc., and a variety of failure sources such as human error, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Hardware failures were the single largest root cause of telecom security incidents in the EU in the year 2021 [43].	Medium
Radio Interference	Radio Interference (AP7)	A DoS attack utilizing radio interference, e.g. by means of jamming or by using compromised devices, may deteriorate the radio signal quality and therefore lead to reduced data rates experienced by the user or even to a complete loss of connection. Services like instant messaging or voice telephony have low data rate requirements and are therefore generally robust against the presence of radio interference. Any DoS attack utilizing radio interference would therefore require to entirely disable one or more radio cells to cause connection failure. The effort to conduct such attacks is high with respect to the achievable gain and impacted geographical area. Therefore, such attacks are rated having a LOW probability.	Low

5.3.8 Tapping or Altering of the Geographical Position

The RAN supports location-based services by enabling the computation of position estimates of a target UE at a location server [64]. For that purpose, the RAN broadcasts assistance information received from the core network, conducts measurements for the target UE and provides the respective measurement information to the core network. The RAN itself has only coarse information about the UE position in the form of the serving cell identifier, i.e. by knowing which cell does serve the target UE. Further, the RAN is not aware of the identity, i.e., SUPI, of the target UE for which the position shall be determined.

Altering of the geographical position may occur if the best serving cell(s) of the target UE are in outage, e.g. as a result from a targeted attack. In such case, the target UE may be served by a suboptimum cell, possibly further away from the actual position of the target UE. As a result, the serving cell identifier may not provide a good position estimate of the target UE, and precise positioning algorithms, e.g. algorithms based on triangularization, may neither provide a good position estimate of the target UE.

So, the **RAN is considered** a relevant attack domain for the risk scenario of Tapping or Altering of the Geographical Position.

The assessment is confined to the case where the user position is determined by the network, particularly by means of the serving cell identifier of the target UE. Alternatively, the user position may be determined on the user side, e.g. by means of GNSS, or by other methods.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	As the number of RAN parameters with impact on positioning performance (e.g. configuration of positioning reference signals) is limited, the probability of RAN misconfiguration to cause tapping or altering of the geographical position is rated as LOW .	Low
Insufficient Access Controls	Availability (AP2-A)	An adversary with remote or physical access to the RAN may be able to modify the positioning measurement information provided to the core network or the positioning assistance information received from the core network. As such an attack would require high effort and sophistication and bring little gain, the probability of insufficient access control to cause tapping or altering of the geographical position is rated as LOW .	Low
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws may concern the performance of the algorithms implemented in the base stations to compute the positioning information, such as measurements of time delays based on the signals received by the base station from the target UE. As the complexity of these algorithms appears limited, the probability of implementation flaws to cause tapping or altering of the geographical position is rated as LOW .	Low
Specification Weakness	Availability (AP4-A)	There are some specification weaknesses which may be exploited to impair or disable 5G radio cells, for example by manipulating the Master Information Block (MIB) to cause cell barring [39]. In turn, such an attack may impair positioning accuracy, particularly if the identifier of the serving cell is used as location information. The effort to conduct such attacks is high with respect to the achievable gain and impacted geographical area. Therefore, such attacks are rated having a LOW probability.	Low
Failure of Power Supply	Availability (AP5-A)	Failure of power supply may result in the unavailability of the best serving cell(s) for the target UE and may therefore impair positioning accuracy, particularly if the identifier of the serving cell is used as location information. Power outage may be caused by human error, power system failure, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Given the variety of different sources for failures of power supply, their probability to be the cause of tapping or altering of the geographical position is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	Failure of hardware components may result in the unavailability of the best serving cell(s) for the target UE and may therefore impair positioning accuracy, particularly if the identifier of the serving cell is used as location information. The probability of a hardware failure to cause tapping or altering of the geographical position is rated as MEDIUM, as there are a wide variety of components that may cause the failure, such as failed Remote (RU), Distributed (DU) or Centralized Units (CU), routing or switching equipment, cable breaks, physical damage of connectors, masts, etc., and a variety of failure sources such as human error, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Hardware failures were the single largest root cause of telecom security incidents in the EU in the year 2021 [43].	Medium
Radio Interference	Radio Interference (AP7)	A DoS attack utilizing radio interference, e.g. by means of jamming or by using compromised devices, may cause the target UE to select a suboptimum serving cell (e.g. a cell further away from the target UE) and therefore lead to impaired positioning accuracy, particularly if the identifier of the serving cell is used as location information. The effort to conduct such attacks is high with respect to the achievable gain and impacted geographical area. Therefore, such attacks are rated having a LOW probability.	Low

5.4 Private 5G Network Operator Perspective

This assessment assumes a private campus network with typical network and security architectures as described in e.g. [65] and [66], respectively. Private mobile networks are often deployed to improve the efficiency of the operations conducted on the campus, e.g. manufacturing facilities, logistics centers, mines, harbors, airports, railway yards, etc. Private networks may also be deployed at public sites such as hospitals, train stations, or exhibition centers.

Depending on the deployment, both RAN and core network may be hosted on the campus. Organizations may also operate multiple locations and in such case prefer the deployment of centralized core network functions and keep user plane traffic locally on campus. The network is typically separated by a firewall, which is part of the private campus network, from the public network, and the network's backbone is part of the organization's general IT network. The members of the organization operating the network may also be the users of the network. In this case, the network may be managed by operations teams having their primary skills in Information (IT) or Operational Technology (OT) domains, rather than in cellular communications. Alternatively, the network may be operated by a specialized operator, separate from the actual users of the network.

For private wide area networks, e.g. for energy utilities, public safety, or the railways, the assessment of Section 5.2 applies, as private wide area networks have good communality with public MNO networks.

5.4.1 Network Failure

The protection level for entering the locations where base stations are physically located could vary a lot. In closed factory facilities there may be a very high protection level. In publicly accessible buildings (e.g. hospitals or stations) the protection level may be low. If an attacker could get physical access to the desired base stations, the attacker could easily make them unusable. That would lead to a non-availability of the network in a small area.

So, the **RAN is considered** a relevant attack domain for the risk scenario for network failure.

For a private 5G network it is assumed that the risk scenario of a network failure relates to the outage of the entire RAN, while the risk scenario of degraded services/connections may be caused by partial outage of the RAN.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	A private RAN deployment provides a wide range of configuration parameters for the Remote (RU), Distributed (DU), and Centralized Units (CU) and for the RAN transport network elements that need to be managed. The private network may be managed by a specialized external network operator, resulting in a low probability of a misconfiguration causing network failure. In other deployment scenarios, external network operators may be avoided, e.g. for fear of espionage at manufacturing sites, and the network configuration may be under control of operations teams having their primary skills in IT or OT systems rather than in cellular communications. Misconfiguration can be expected primarily to cause impaired network performance and less likely an entire network failure. Concluding from the above, the probability of a misconfiguration causing network failure is rated as MEDIUM .	Medium
Insufficient Access Controls	Availability (AP2-A)	The radio sites of a private network are often well protected and surveyed, e.g. if deployed at manufacturing sites. Deployment at public places such as stations or hospitals eases the physical access to the radio sites. This simplifies the destruction of one or multiple cell sites by targeted attacks or vandalism. To cause a complete network outage it is expected to require significant effort and sophistication, e.g. a cyber-attack to gain access to the RAN management system. A cyber attacker can primarily be expected to have objectives other than shutting down a private network, or in such case may prefer to attack the core or transport network. Therefore, the probability of insufficient access control to cause network failure by attacking the RAN domain is rated as LOW .	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered as having MEDIUM probability to cause impairment or failure of a private RAN, as such flaws may have a variety of causes and may be non-trivial to detect, for example: - SW upgrades can lead to the use of flawed or misconfigured SW [24][25]. - The implementation may be compromised, e.g. compromised open-source SW may be used, particularly if using a cloud platform [14][15][74][74]. - The RAN includes a variety of performance impacting algorithms and features that may be flawed, particularly as low-cost product variants may be preferred for private networks. Faulty SW updates were the single largest cause of telecom user hours lost in the EU in year 2022 [10].	Medium
Specification Weakness	Availability (AP4-A)	Exploitation of specification weaknesses of the 5G RAN to implement DoS attacks, e.g. by altering the Master Information Block (MIB) to indicate cell barring [39], is rated having a LOW probability to cause network failure, as such attacks would require high effort and sophistication.	Low
Failure of Power Supply	Availability (AP5-A)	Failure of a private RAN may be caused by regional or localized power outage. Power outage may be caused by human error, power system failure, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Power outage may likely affect also other components of the private network or lead to a shut-down of the entire operation on the site (e.g. manufacturing site) on which network is deployed. To reduce the risk of a failure of power supply, a private network operator will likely use an uninterruptable power supply (UPS). Nevertheless an UPS in a private 5G network is probably smaller in size in comparison to one in a public 5G network. Given the variety of different sources for failures of power supply, their probability to cause network failure is rated as HIGH .	High
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to cause impairment or failure of the RAN is rated as HIGH , as there are a wide variety of components that may cause the failure, such as failed Remote (RU), Distributed (DU) or Centralized Units (CU), routing or switching equipment, cable breaks, physical damage of connectors, masts, etc., and a variety of failure sources such as human error, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Private networks may also be deployed without a good level of redundancy which may cause the entire network to fail if a hardware failure occurs at a centralized location in the network, for example, an aggregation router/switch or a server hosting a cloudified CU.	High
Radio Interference	Radio Interference (AP7)	Failure of a private network may be caused by a localized DoS attack utilizing radio interference, e.g. by means of jamming the cellular air interface or the GNSS receiver providing the master clock. Such a DoS attack can cause large financial damage, e.g. if operations at a manufacturing site get impaired, and can be performed with limited effort. Therefore, the probability of such an attack to cause network failure is rated as MEDIUM .	Medium

5.4.2 Degraded Services / Connections

As described in Chapter 5.4.1, for a private 5G network it is assumed that the risk scenario of a network failure relates to the outage of the entire RAN, while the risk scenario of degraded services/connections may be caused by partial outage of the RAN. This happens for example if an attacker could get physical access to any part of the coverage area of the desired base station and place an electronic device creating radio interfering signals. As a result, the service quality can be degraded within a rather limited area.

So, the **RAN is considered** a relevant attack domain for the risk scenario of degraded services / connections.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Availability (AP1-A)	Due to the complexity of the air interface and network elements of the 5G System, a private RAN deployment provides a wide range of configuration parameters for the Remote (RU), Distributed (DU), and Centralized Units (CU) and for the RAN transport network elements that have impact on network performance and need to be managed properly. Misconfiguration can be expected to more likely cause impaired network performance than an entire network failure. The private network may be managed by a specialized external network operator. For this deployment scenario, the probability of a misconfiguration causing impaired network performance is rated as medium. In other deployment scenarios, external network operators may be avoided, e.g. for fear of espionage at manufacturing sites, and the network configuration may be under control of operations teams having their primary skills in IT or OT systems rather than in cellular communications. For those scenarios, the probability of a misconfiguration causing degraded services / connections is rated as medium-to-high. According to the maximum principle, this scenario is therefore rated as HIGH .	High
Insufficient Access Controls	Availability (AP2-A)	The radio sites of a private network are often well protected and surveyed, e.g. if deployed at manufacturing sites. Deployment at public places such as stations or hospitals eases the physical access to the radio sites. This simplifies the destruction of one or multiple cell sites by targeted attacks or vandalism which in turn results in impaired network performance. Therefore, the probability of insufficient RAN access control to cause degraded services / connections is rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Implementation Flaw (Quality of the Product)	Availability (AP3-A)	Implementation flaws are considered having a MEDIUM probability to cause impairment or failure of a private RAN, as such flaws may have a variety of causes and may be non-trivial to detect, for example: - SW upgrades can lead to the use of flawed or misconfigured SW [24][25]. - The implementation may be compromised, e.g. compromised open-source SW may be used, particularly if using a cloud platform [14][15][74][74]. - The RAN includes a variety of performance impacting algorithms and features that may be flawed, particularly as low-cost product variants may be preferred for private networks. Faulty SW updates were the single largest cause of telecom user hours lost in the EU in the year 2022 [10].	Medium
Specification Weakness	Availability (AP4-A)	Exploitation of specification weaknesses of the 5G RAN to implement DoS attacks, e.g. by altering the Master Information Block (MIB) to indicate cell barring [39], attacks exploiting 5G particularities (e.g. to cause frequent RRC state transitions [6]) or using compromised UEs emulating high priority users [9], is rated having a LOW probability to cause degraded services / connections, as such attacks would require high effort and sophistication.	Low
Failure of Power Supply	Availability (AP5-A)	Performance impairment of a private RAN, e.g. by disabling one or several radio sites, may be caused by localized power outage. Power outage may be caused by human error, power system failure, a natural disaster, sabotage, etc. To reduce this risk, a private network operator will likely use an UPS. Nevertheless an UPS in a private 5G network is probably smaller in size in comparison to one in a public 5G network. On the other hand, the likelihood of a failure of power supply in a private network leading to a complete network failure can be assumed higher, than leading to a degradation of services. Taking this and the variety of different sources for localized failures of power supply into account, their probability to cause degraded services / connections is rated as MEDIUM .	Medium
Failure of Hardware Components or Unavailability of Spare Parts	Availability (AP6-A)	The probability of a hardware failure to cause RAN performance degradation is rated as HIGH , as there are a wide variety of components that may cause the failure, such as failed Remote (RU), Distributed (DU) or Centralized Units (CU), routing or switching equipment, cable breaks, physical damage of connectors, masts, etc., and a variety of failure sources such as human error, a natural disaster, a terror attack, e.g. [40], or sabotage or theft of components, e.g. [41]. Private networks may also be deployed without a good level of redundancy which may cause a degradation of services if a hardware failure occurs at a centralized location in the network, for example, an aggregation router/switch or a server hosting a cloudified CU.	High

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Radio Interference	Radio Interference (AP7)	Performance impairment of a private network may be caused by a localized DoS attack utilizing radio interference, e.g. by means of jamming the cellular air interface or the GNSS receiver providing the master clock. Such DoS attack can cause large financial damage, e.g. if operations at a manufacturing site get impaired, and can be performed with limited effort. Therefore, the probability of such an attack to cause degraded services / connections is rated as MEDIUM .	Medium

5.4.3 Information Disclosure

There are potential misconfigurations or implementation flaws that could lead to a deactivation of encryption or to the use of weak encryption at the air or network interfaces.

So, the **RAN is considered** a relevant attack domain for the risk scenario for information disclosure.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	In the case that user plane ciphering is not enabled, an attacker could extract sensitive user or business information. As the configuration and transport layer and application layer encryption heavily depend on the security concepts of the corresponding company, encryption might not be enforced or could be bypassed. Further, the correct configuration of the RAN depends on the knowledge of the respective operations team that might not have the same expertise as an MNO. The probability is therefore rated as MEDIUM .	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Confidentiality (AP2-C)	If an attacker can compromise credentials of cloud components, the attacker might be able to access other components horizontally, but also vertically, depending on the privileges. This could enable an attacker to extract sensitive information about business processes and other business information or data of employees. Furthermore, if resources are shared within or between systems and are not adequately protected by access controls, sensitive information can also be extracted from there. When it comes to physical attacks, radio sites of private networks are often well protected. Additionally, it should be considered, that security personnel are often deployed to improve physical access controls. In contrast, deployments at public places such as stations or hospitals ease the physical access to radio sites. However, to extract sensitive information, attacks require a high level of effort. Depending on the security measures installed, the probability for attacks in this scenario can be rated between low and medium. According to the maximum principle, this scenario is therefore rated as MEDIUM.	Medium
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	Software updates can affect the configuration parameters and introduce new parameters with corresponding new default values. Updates may also overwrite existing parameters. In both cases, this could lead to incorrect configuration of the RAN components, which means that user data is not adequately protected. However, software updates can be assumed to be well tested and changes to existing parameters can be assumed to occur rarely. Access to keys and identities through implementation flaws can allow an attacker to track users. This might not be very beneficial as the identities would need to be correlated with personal data (such as employee-specific information) from outside the RAN to be useful for a user mapping or for further data extractions. Further, such an access can for instance be utilized through supply chain attacks that contain cryptographic backdoors with weakened cryptography. As attacks on opensource software become more common and more sophisticated, vendors have to develop a development and deployment strategy. Additionally, the update lifecycle of vendor-specific soft- and hardware heavily depends on the management of the private operator. The probability will therefore be rated as MEDIUM.	Medium
Specification Weakness	Confidentiality (AP4-C)	User priority is transmitted on the air interface without encryption, which allows attackers to track groups of priority users, but not individuals specifically, which in most situations provides little benefit to attackers, as exploitation is associated with medium effort. The probability is rated LOW, because attackers can track priority users with medium effort while private 5G operators might only have few or no use cases for using priority groups.	Low

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Failure of Power Supply	Confidentiality (AP5-C)	The failure of power supply appears unrelated to information disclosure.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	Hardware failures appear unrelated to information disclosure.	Low
Radio Interference	Radio Interference (AP7)	Attacks utilizing radio interference appear unrelated to information disclosure.	Low

5.4.4 User Identities (Authenticity/Integrity)

The integrity of messages at the air interface may not be verified so that an attacker may be able to insert messages that are pretended to be sent from the true sender.

So, the **RAN is considered** a relevant attack domain for the risk scenario for user identities.

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Misconfiguration	Confidentiality (AP1-C)	Encryption and integrity protection between the RAN and the UE might be disabled on the air or network interfaces by misconfiguration, which enables attackers to extract sensitive user or business information. As the configuration and transport layer and application layer protections heavily depend on the security concepts of the corresponding company, encryption might not be enforced or could be bypassed and a lack of integrity protection can enable attackers to spoof information. The correct configuration of the RAN depends on the knowledge of the respective operation team that might not have the same expertise as an MNO. The probability is therefore rated as MEDIUM.	Medium

Attack Path Category	Attack Path ID	Probability Description	Probability Level
Insufficient Access Controls	Confidentiality (AP2-C)	Insufficient access control to RAN components, their software or shared resources in the RAN could enable an attacker to access sensitive information that might be helpful to extract the user's identity. As RAN components hold only limited user-specific information the probability is rated as LOW.	Low
Implementation Flaw (Quality of the Product)	Confidentiality (AP3-C)	An attacker can include backdoors on cryptographic algorithms, keys or procedures via supply chain attacks. This could be used to weaken the cryptography which could then allow attackers to extract information about the SUPI or the user. Further, infrequent updates of temporary user identifiers might enable attackers to track a user's position [11]. As these attacks require high effort and provide low benefit, the probability is therefore rated as LOW.	Low
Specification Weakness	Confidentiality (AP4-C)	User priority is transmitted on the air interface without encryption, which allows attackers to track groups of priority users [9], but not individuals specifically, which in most situations provides little benefit to attackers. Further, unprotected data before authentication could be used to fingerprint UEs. The probability is rated as LOW, because attackers can track priority users (e.g. police movements) with medium effort while private 5G operators might only have few or no use cases for using priority groups.	Low
Failure of Power Supply	Confidentiality (AP5-C)	The failure of power supply appears unrelated to the exposure of a user's identity.	Low
Failure of Hardware Components or Unavailability of Spare Parts	Confidentiality (AP6-C)	The failure of hardware components or unavailability of spare parts appears unrelated to the exposure of a user's identity.	Low
Radio Interference	Radio Interference (AP7)	Radio interference appears unrelated to the exposure of a user's identity.	Low

6 Risk Assessment Results

This chapter presents the final results of the detailed risk analysis.

Note: Risk scenarios with no given probability level were not rated, see beginning of Chapter 5 and the corresponding risk scenario.

6.1 Probability Assessment

The following table summarizes the results of this detailed analysis of the **Radio Access Network (RAN)**. The final risk analysis that covers all subsystems is presented in the framework document.

Perspective	Section	Risk Scenario	Probability Level
Public MNO	5.2.1	Emergency Call (TKG § 164)	
	5.2.2	Cell Broadcast (TKG § 164a)	High
	5.2.3	Telecommunications Secrecy (TTDSG § 3)	Medium
	5.2.4	Emergency Preparedness (TKG § 184)	Medium
	5.2.5	Regional Network Failure	High
	5.2.6	Multi-Regional Network Failure	
	5.2.7	Failure of Mobile Service	
	5.2.8	Fraud	
	5.2.9	Subscriber-Data Leak	Medium
User	5.3.1	Urgent Telephone Calls Not Possible	Medium
	5.3.2	Authenticity of the Receiver of a Call	
	5.3.3	Integrity of Transmitted Data	Medium
	5.3.4	User's Identity	Low
	5.3.5	Eavesdropping of a Communication	Medium
	5.3.6	Slowing Down/Restricting the Connection	Medium
	5.3.7	Connection Failure	Medium
	5.3.8	Tapping or Altering of the Geographical Position	Medium
Private 5G Network Operator	5.4.1	Network Failure	High
	5.4.2	Degraded Services/Connections	High
	5.4.3	Information Disclosure	Medium
	5.4.4	User Identities (Authenticity/ Integrity)	Medium

6.2 Risk Assessment

The following table summarizes the overall risk assessment of the **Radio Access Network (RAN)**. The risk scenarios and their respective impact level are described in the framework document.

Perspective	Risk Scenario	Impact	Probability	Risk Score
Public MNO	Emergency Call (TKG § 164)			
	Cell Broadcast (TKG § 164a)	High	High	High
	Telecommunications Secrecy (TTDSG § 3)	Medium	Medium	Medium
	Emergency Preparedness (TKG § 184)	High	Medium	High
	Regional Network Failure	Medium	High	High
	Multi-Regional Network Failure			
	Failure of Mobile Service			
	Fraud			
	Subscriber-Data Leak	Medium	Medium	Medium
User	Urgent Telephone Calls Not Possible	Low	Medium	Low
	Authenticity of the Receiver of a Call			
	Integrity of Transmitted Data	High	Medium	High
	User's Identity	High	Low	Medium
	Eavesdropping of a Communication	Medium	Medium	Medium
	Slowing Down/Restricting the Connection	Low	Medium	Low
	Connection Failure	Low	Medium	Low
	Tapping or Altering of the Geographical Position	High	Medium	High
Private 5G Network Operator	Network Failure	Medium	High	High
	Degraded Services/Connections	Medium	High	High
	Information Disclosure	Medium	Medium	Medium
	User Identities (Authenticity/ Integrity)	Medium	Medium	Medium

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